

The Challenge of Preserving Agricultural Capacity

The United States has built one of the most productive agricultural systems in the world. Over the years, thanks to advances in science, technology, and management, we've significantly increased our agricultural output. This growth has not only supported our expanding population but has also made a big impact on the global food supply.

However, just because we've been successful in the past doesn't mean we can expect the same results in the future. Agricultural systems are constantly influenced by changing environmental, economic, and technological factors. Issues like population growth, shifting consumer preferences, new pests and diseases, resource scarcity, technological changes, workforce dynamics, global political shifts, and climate variability all play a role in shaping our agricultural capabilities. For policymakers, farmers, researchers, and the agricultural community as a whole, the challenge isn't just to maintain our current production levels; it's also about ensuring we can adapt to these evolving conditions in the coming years.

It's important to recognize the difference between production and capacity. Production refers to the quantities of food, fibers, fuels, and other products we generate over a certain period.

Capacity, on the other hand, is about the long-term ability of our agricultural system to keep producing or even improve over time, despite whatever challenges may arise. Capacity includes not only what we're producing now but also the support systems that allow us to continue, innovate, and adapt.

To grasp these underlying systems, we need to think long-term. Many of the institutions and resources that keep our agricultural sector thriving have been built up over decades through ongoing investments. Similarly, many of the challenges we face today, like resource depletion and changing demographics, have developed gradually and will affect agriculture for a long time.

This report takes a long-term view. Instead of focusing on specific agricultural programs or policies, it looks at the broader scientific and economic evidence that has historically helped the U.S. maintain a strong and adaptable agricultural system. The goal is to pinpoint the conditions most closely linked with our agricultural capacity and understand their importance for the future, spanning the next 25 to 50 years.

Principal Findings

Finding 1 — Sustained improvements in total factor productivity have been the principal source of long-run growth in U.S. agricultural output.

High-confidence evidence indicates that post-war agricultural production expanded primarily through improvements in productivity rather than through proportional increases in land, labor, or other conventional production inputs. This historical trend provides the empirical foundation for understanding the long-term capacity of U.S. agriculture.

Finding 2 — Productivity growth has resulted from continuous innovation rather than isolated technological breakthroughs.

The historical record demonstrates that successive advances in biological science, engineering, farm management, and information systems have collectively increased agricultural productivity

over multiple decades. Long-term productivity growth reflects the cumulative effects of scientific discovery, technological development, and producer adoption.

Finding 3 — Agricultural capacity depends upon the interaction of multiple reinforcing systems.

No single factor explains the resilience of American agriculture. Scientific research, technological innovation, human capital, infrastructure, economically viable producers, effective institutions, and productive natural resources operate as an interconnected system. Weakness in any one component may reduce the performance of the system as a whole.

Finding 4 — Natural resources constitute productive capital rather than passive environmental assets.

Healthy soils, reliable water supplies, biodiversity, and ecosystem functions provide the biophysical foundation upon which agricultural productivity ultimately depends. Long-term agricultural capacity cannot be evaluated independently of the condition of these underlying resources.

Finding 5 — Adaptation has been a defining characteristic of U.S. agricultural success.

The historical record demonstrates repeated adaptation to changing technologies, environmental conditions, markets, pests, diseases, and consumer demands. The ability to generate new knowledge and incorporate it into agricultural practice has been a defining feature of the American agricultural system.

Finding 6 — Preserving agricultural capacity requires maintaining the conditions that enable continued adaptation.

The evidence reviewed in this assessment indicates that long-term agricultural resilience depends

not only on preserving current productive capacity but also on sustaining the scientific, institutional, environmental, and economic systems that generate future improvements in productivity.

These findings provide the analytical framework for the remainder of the report. Subsequent chapters evaluate each finding in greater detail, assess the strength of the supporting evidence, identify important uncertainties, and examine their implications for preserving the productive capacity of United States agriculture over the next 25 to 50 years.

Research Question

This assessment is guided by a single overarching question:

What has enabled the United States to build and sustain one of the world's most productive and resilient agricultural systems, and what will be required to preserve that strength over the next 25 to 50 years?

Rather than focusing on any single challenge or policy issue, the report examines the broader conditions that have historically supported agricultural success and considers how those conditions can be maintained in an era of environmental, technological, economic, and demographic change.

Scope

This assessment examines the long-term productive capacity of United States agriculture rather than short-term changes in commodity prices, annual production levels, or temporary market

conditions. Its purpose is to identify the enduring factors that have allowed American agriculture to remain productive, innovative, and adaptable over time.

The analysis draws upon evidence from agricultural economics, agronomy, environmental science, engineering, rural sociology, public policy, and related fields to evaluate broad national trends rather than individual farms, regions, or industries.

Six interconnected dimensions of agricultural capacity are considered throughout the report:

- Innovation capacity, including scientific research, technological advancement, and the adoption of improved production practices.
- Natural capital, including soils, water resources, biodiversity, and the ecological systems that make agricultural production possible.
- Human capital, including the education, technical expertise, and workforce needed to sustain agricultural productivity and innovation.
- Physical infrastructure, including transportation networks, irrigation systems, energy, communications, storage, and other essential supporting infrastructure.
- Institutional capacity, including the organizations, governance systems, research institutions, and information networks that enable agricultural adaptation and long-term planning.
- Economic resilience, including the financial conditions that allow producers to maintain productive assets, manage risk, and invest in future productivity.

Although each of these areas is examined individually, the report emphasizes that they function as parts of an integrated system. The long-term performance of American agriculture depends not on any single component, but on the strength of the relationships among them.

Scope Limitations

Several important topics fall outside the purpose of this assessment.

The report does not attempt to forecast commodity prices, predict future production levels, evaluate individual federal or state agricultural programs, or advocate for specific legislative proposals. Nor is it intended to serve as a comprehensive history of American agriculture.

These subjects are discussed only when they help explain the broader question of how the United States can preserve and strengthen its agricultural capacity over the coming decades.

Organization of the Report

The report begins by examining the historical forces that shaped the remarkable growth of agricultural productivity in the United States. It then explores the major systems that have sustained that progress, including scientific innovation, natural resources, human capital, infrastructure, institutions, and economic resilience.

Building on this foundation, the later chapters identify the long-term trends most likely to influence American agriculture over the next half century and develop a strategic framework for preserving agricultural capacity. The report concludes with evidence-based recommendations intended to strengthen the nation's ability to remain productive, resilient, and adaptable in the decades ahead.

The Historical Record

Over the past several decades, the United States has achieved extraordinary gains in agricultural productivity. American farms now produce substantially more agricultural output than they did in the middle of the twentieth century, even though aggregate use of major production inputs has changed relatively little. USDA Economic Research Service (ERS) productivity accounts show that between 1948 and 2023, farm output grew at an average annual rate of 1.52 percent, while aggregate input use grew by only 0.12 percent per year. Total factor productivity (TFP), which captures output growth not explained by growth in measured inputs, increased at an average annual rate of 1.40 percent over the same period (U.S. Department of Agriculture, Economic Research Service [USDA ERS], 2026).

This pattern is important because it changes how the historical development of American agriculture should be understood. The sector did not become substantially more productive simply by bringing more land into production or employing more workers. Instead, producers found increasingly effective ways to use the resources already available to them. Advances in genetics, machinery, chemicals, management, farm organization, and other technologies helped increase the efficiency with which agricultural inputs were converted into output (Wang et al., 2015; USDA ERS, 2026).

The distinction between input growth and productivity growth provides an important foundation for this assessment. If long-term agricultural growth had depended primarily on expanding the quantity of land, labor, and other conventional inputs, preserving future capacity would largely require maintaining access to those resources. The historical record points somewhere else. Continued productivity improvement has been the principal means through which U.S. agriculture has expanded output while limiting the growth of conventional inputs (Heisey, Wang, & Fuglie, 2012; USDA ERS, 2026).

The central question, then, is not simply how much American agriculture can produce today. It is how the agricultural system has repeatedly become more capable of producing from the resources available to it—and whether the institutions that generated those gains can continue doing so.

Evidence for Productivity-Led Growth

The historical productivity record provides a relatively straightforward answer to one of the central questions of this assessment: How did American agriculture produce substantially more without a comparable increase in the resources used to produce it?

The answer is productivity growth. USDA ERS estimates that U.S. farm output more than tripled relative to its 1948 level by 2023, while aggregate input use increased only modestly. The resulting growth in TFP accounted for most of the increase in output over the period (USDA ERS, 2026). Earlier analyses reaching back to 1948 reached the same basic conclusion: productivity rather than expanding input use was the dominant source of long-run agricultural growth (Wang et al., 2015).

TFP is particularly useful for this assessment because it captures improvements that are otherwise difficult to assign to a single input. Changes reflected in TFP can include technological advances, improvements in the performance of on-farm activities, changes in farm organization and structure, improvements in crop and animal genetics, and other forms of technical change (USDA ERS, 2026).

This helps explain why the historical productivity record is more than a collection of output statistics. It reflects a broader process of learning and improvement. Scientific discoveries,

technological development, improved management, and changes in how agricultural businesses are organized have allowed producers to obtain progressively greater output from the resources available to them (Wang et al., 2015; USDA ERS, 2026).

The gains have also accumulated over long periods rather than appearing as the result of a single breakthrough. Earlier USDA analyses found that productivity contributed positively to agricultural output across successive periods since 1948, with innovations involving crop and animal genetics, chemicals, equipment, and farm organization contributing to those gains (Wang, 2020). More recent ERS estimates continue to identify technical change as the principal driver of long-term TFP growth (USDA ERS, 2026).

The historical lesson is therefore broader than simply saying that productivity increased.

Long-run agricultural growth has depended primarily on the ability to keep improving productivity. Preserving agricultural capacity in the future will consequently require maintaining the research, education, infrastructure, and institutional systems that make continued productivity gains possible (USDA ERS, 2025, 2026).

From Productivity to Innovation

The productivity record establishes what happened. The next question is more fundamental:

What made those gains possible?

TFP statistics cannot answer that question by themselves. They show that agriculture became more efficient, but the measure combines the effects of many different changes. Understanding the forces behind productivity growth therefore requires looking beyond the aggregate

productivity measure to the scientific, technological, institutional, and economic systems that produced those changes (USDA ERS, 2026).

The evidence points to a process of continuous innovation rather than a succession of isolated breakthroughs. Agricultural productivity has benefited from advances in genetics, engineering, agronomy, chemistry, information technology, farm management, and other fields. Research on U.S. agricultural productivity identifies technical change, fueled in large part by public and private research and development, as a principal source of long-term TFP growth (Alston et al., 2009, 2010; Fuglie & Heisey, 2007; Huffman & Evenson, 2006; Plastina & Fulginiti, 2012; Wang et al., 2013).

Innovation also involves much more than inventing something entirely new. Improvements can come from refining existing technologies, changing management practices, reallocating resources more effectively, or adapting a technology to different environmental and economic conditions. USDA ERS's productivity framework explicitly recognizes improvements in on-farm practices, farm organization, crop and animal genetics, and other technical changes as contributors to TFP growth (USDA ERS, 2026).

The distinction between productivity and innovation is therefore important. Productivity is the outcome; innovation is one of the processes that produces it. Researchers generate knowledge, engineers and firms turn knowledge into practical tools, producers test and modify those tools, and successful practices spread through the agricultural economy. Over time, the combined effects of those activities become visible in productivity statistics.

This perspective also explains why agricultural innovation is better understood as a system than as an inventory of technologies. Research, education, infrastructure, Extension, markets, and

producer experience all influence whether a new idea can move from discovery to widespread use (Wang et al., 2012; Wang et al., 2015; USDA ERS, 2026).

The Agricultural Innovation System

The development of American agriculture is best understood as a continuing cycle of discovery, experimentation, adoption, and refinement. Scientific research generates new knowledge; researchers, engineers, and firms turn that knowledge into practical technologies; producers evaluate those technologies under real production conditions; and the experience gained through adoption can inform subsequent research and development. This reciprocal process has been an important feature of agricultural productivity growth (Wang et al., 2012, 2013; USDA ERS, 2026).

Innovation in agriculture is therefore inherently cumulative. Scientific discoveries often require substantial periods of testing and development before they become commercially useful, and even successful technologies may require further refinement after they reach farms and ranches. Improvements in reliability, cost, efficiency, and compatibility with existing production systems can determine whether a promising technology becomes widely adopted (USDA ERS, 2026; Wang et al., 2015).

The process also depends on a diverse institutional network. Public research organizations have historically contributed foundational knowledge and research addressing broad social and environmental objectives. Private firms have played a major role in developing and commercializing marketable technologies, particularly in areas such as agricultural machinery and other inputs. ERS research indicates that public and private agricultural research are often

complementary rather than substitutes, with their relative roles varying across areas of research (Fuglie & Heisey, 2007; USDA ERS, 2019).

This complementarity is important because research with broad or uncertain social benefits may not generate sufficient private incentives on its own. Public agricultural research has historically produced benefits extending beyond individual firms or commodities, while private research provides incentives for developing commercially viable products and technologies. Empirical research has found substantial economic returns to public agricultural research and a strong relationship between public agricultural R&D and productivity growth (Fuglie & Heisey, 2007; Heisey, King, & Day-Rubenstein, 2011; Heisey, Wang, & Fuglie, 2012).

Producers themselves are also part of the innovation system. Technologies and management practices are rarely adopted in precisely the form in which they were originally developed. Farmers and ranchers adapt them to local soils, weather, markets, labor conditions, and operational constraints. Those adaptations can improve the usefulness of existing technologies and provide practical information that feeds back into future research and development (Wang et al., 2012; Wang et al., 2015).

Agricultural innovation is consequently best understood as a network of institutions and people learning from one another. Its strength lies not only in generating new ideas, but also in the ability to test, modify, communicate, and apply those ideas across a diverse agricultural economy.

Innovation as the Engine of Productivity

The historical record establishes a clear relationship between productivity and innovation. Long-run TFP growth is driven primarily by technical change, and the literature identifies research and development as one of the principal sources of that technical change (USDA ERS, 2026; Alston et al., 2009, 2010; Wang et al., 2013).

The relationship is broader than technological invention alone. Improvements in farm organization, management, genetics, resource allocation, and production practices can all contribute to productivity growth. In this sense, innovation includes both major technological advances and the countless smaller decisions through which producers refine the way agricultural resources are used (USDA ERS, 2026; Wang et al., 2015).

That broader definition matters for agricultural capacity. A productive agricultural system is not simply one that possesses advanced technologies. It is one that can continually generate or obtain useful knowledge, evaluate new approaches, learn from experience, and incorporate improvements into production. These capabilities give producers more ways to respond when environmental, economic, or technological conditions change.

Innovation is therefore both a source of productivity and an element of resilience. Research and technology can improve efficiency under current conditions while also expanding the options available to producers when those conditions change.

Agricultural Research and Development

Agricultural research and development sits at the heart of this process. Research rarely increases agricultural production immediately. Its more important contribution is to expand the stock of knowledge from which improved crop varieties, livestock genetics, production practices,

engineering systems, pest-management strategies, and decision-support tools can emerge (Fuglie & Heisey, 2007; USDA ERS, 2025, 2026).

Research is consequently a long-term investment. Scientific discoveries must often pass through stages of experimentation, validation, development, and adoption before their effects become visible in national productivity measures. The returns may also persist after adoption as technologies are refined and applied in new settings. Economic research has consistently found high returns to agricultural research investment, although the timing and magnitude of individual research benefits vary substantially across projects and fields (Fuglie & Heisey, 2007; Heisey, King, & Day-Rubenstein, 2011).

The history of American agriculture also demonstrates the importance of research across multiple disciplines. Plant breeding, animal genetics, soil science, agronomy, engineering, veterinary medicine, chemistry, and information technologies have each contributed to improvements in agricultural productivity. USDA ERS identifies innovations in crop and animal genetics, farm organization, machinery, chemicals, and other technologies as important contributors to long-term productivity growth (USDA ERS, 2026; Wang et al., 2015).

Research also expands the range of responses available when agriculture faces new problems. Improvements in genetics, pest and disease management, water-use efficiency, and decision-support technologies can help producers respond to environmental and biological pressures while continuing to pursue productivity gains. The value of research therefore extends beyond the immediate technology it produces; it also increases the ability of the agricultural system to respond to problems that may not yet be fully understood.

Translating Research into Agricultural Innovation

Scientific knowledge matters only when it can be translated into practice. A discovery can contribute to agricultural productivity only if it eventually becomes a technology, production practice, management approach, or other tool that producers can use under real operating conditions. This translation process is therefore a critical link between research investment and productivity growth (USDA ERS, 2026).

The process is rarely linear. Research findings may move into applied research, field testing, engineering development, commercial production, and producer evaluation, but information also flows in the opposite direction. Producers can identify practical problems that were not apparent during research, and those observations can influence subsequent research and product development. NIFA explicitly describes the integration of research, education, and Extension as a means of accelerating the generation, translation, and transfer of knowledge into practical applications (National Institute of Food and Agriculture [NIFA], 2025).

The history of agricultural technology illustrates why this process matters. Technologies often become more useful as they are refined for reliability, affordability, local conditions, and ease of use. Their eventual productivity effects therefore depend not simply on the quality of the original invention but on the institutions and producers involved in testing and improving it.

Successful innovation requires organizations capable of supporting each stage of this process. Research institutions create knowledge, private firms and engineers develop practical applications, Extension and other educational organizations help communicate them, and producers determine how well those applications work under diverse conditions (NIFA, 2025; Wang et al., 2012).

Public and Private Agricultural Research

The productivity gains achieved by American agriculture have depended on both public and private research. Public institutions have historically supported foundational science and research whose benefits extend across industries, regions, and generations. Private firms have increasingly invested in technologies and products for which commercial markets provide stronger incentives (Fuglie & Heisey, 2007; USDA ERS, 2019).

Public research is especially important where the benefits of scientific discovery are broad, difficult to appropriate, or unlikely to generate immediate commercial returns. ERS research finds that public investment in agricultural research has produced high economic returns and has contributed to productivity growth through benefits to producers, the food sector, and consumers (Fuglie & Heisey, 2007; Heisey et al., 2011).

Private research, meanwhile, has become particularly important in market-oriented areas of agricultural technology. ERS data show that private agricultural and food R&D investment increased substantially over the period examined, while public agricultural R&D growth was comparatively slower in later decades. The sectors frequently focus on different types of research and therefore can complement one another rather than simply compete (USDA ERS, 2019).

The relationship is therefore best understood as a partnership. Public research can provide scientific foundations and address questions with broad public benefits, while private firms can develop, commercialize, and refine technologies that respond to market demand. Research on U.S. agricultural productivity has found significant relationships among public R&D, private R&D, and long-run productivity growth, reinforcing the importance of maintaining both parts of the research ecosystem (Wang et al., 2013).

Agricultural Extension and Knowledge Transfer

Research is only the beginning of agricultural innovation. New knowledge must reach producers, be interpreted in ways that are useful to them, and be evaluated under the conditions in which agricultural decisions are actually made. The Cooperative Extension System has historically served this role by connecting land-grant universities and research institutions with producers and communities throughout the United States (NIFA, 2025).

Extension is designed to operate in both directions. University and research personnel translate research findings for practical audiences, while local Extension educators work with communities, producers, and other stakeholders to identify problems and provide feedback that can inform future research priorities (NIFA, 2025).

This local component matters because American agriculture encompasses widely different production systems and environmental conditions. A technology or management practice that works well in one setting may require modification elsewhere. Knowledge-transfer institutions can help producers evaluate these differences rather than treating every innovation as universally applicable.

The effectiveness of knowledge transfer also depends on trust, accessibility, and relevance. NIFA describes Extension as a long-standing nationwide educational network designed to bring research-based information to farmers, ranchers, and communities while relying on local relationships to address specific needs (NIFA, 2025).

For this reason, Extension should be viewed as part of the productive infrastructure of agricultural innovation rather than simply as an educational service. ERS research identifies extension, infrastructure, and technology spillovers as factors that can enhance long-term

agricultural productivity alongside research and development investment (Wang et al., 2012; USDA ERS, 2026).

Producer Adoption and Adaptive Management

Innovation affects productivity only when producers are able and willing to incorporate new knowledge into their operations. Adoption is therefore a critical stage in the process through which scientific research becomes measurable agricultural improvement.

Adoption is rarely uniform. Producers operate under different environmental, financial, labor, and market conditions, and technologies can perform differently across those settings. The productivity literature recognizes that technology adoption and the benefits of innovation depend partly on the surrounding economic and institutional environment, including education, extension, infrastructure, incentives, and access to markets (Fuglie & Wang, 2012; Wang et al., 2012).

Economic considerations matter as well. Producers must weigh the expected benefits of an innovation against its purchase and operating costs, financial risks, technical requirements, and compatibility with existing production systems. The decision to adopt is therefore not simply a test of whether a technology works in principle; it is a judgment about whether it works under a particular set of circumstances.

Producers also contribute directly to innovation through adaptation and experimentation. New technologies and practices are often modified after adoption as producers respond to local conditions and accumulated experience. These adjustments can improve performance and

generate practical knowledge that feeds into subsequent research, technology development, and Extension efforts (NIFA, 2025; Wang et al., 2012).

This makes adaptive management an important part of agricultural productivity. Weather, pests, markets, technologies, and other operating conditions change continually, and producers must adjust their decisions accordingly. A resilient agricultural system therefore depends not only on its ability to generate innovations but also on its ability to learn from experience and incorporate those lessons into subsequent decisions.

Synthesis of the Agricultural Innovation System

The evidence reviewed in this chapter points to a straightforward conclusion: agricultural innovation is a system, not a collection of isolated inventions. Research generates knowledge; technology development turns knowledge into practical tools; Extension and other institutions help communicate and adapt those tools; and producers determine how they perform under real production conditions. The productivity gains associated with innovation emerge from the interaction of these activities over time (Wang et al., 2012, 2013; USDA ERS, 2026).

The historical record also shows why this system must be maintained over long periods. Research institutions, scientific expertise, data systems, educational networks, and Extension organizations cannot be rebuilt instantly after years of decline. Agricultural research has long development horizons, and productivity growth depends in part on maintaining the institutional capacity to produce and diffuse new technologies over successive generations (Fuglie & Heisey, 2007; Wang et al., 2015).

The diversity of organizations involved in agricultural innovation is itself an advantage. Federal and state research institutions, land-grant universities, Extension organizations, private firms, producer groups, and individual producers contribute different forms of expertise and create multiple pathways for developing and applying new knowledge. Public and private research therefore function most effectively when their complementary roles are preserved rather than treated as substitutes (USDA ERS, 2019; Wang et al., 2013).

The central implication for this assessment is clear: innovation capacity is itself a component of agricultural capacity. The long-term productivity of American agriculture will depend not only on maintaining productive land and physical infrastructure, but also on preserving the research institutions, educational systems, knowledge networks, and producer relationships that allow the sector to keep learning and improving.

Chapter Findings

Finding 1: Continuous innovation has been a principal driver of long-term productivity growth in United States agriculture (USDA ERS, 2026; Wang et al., 2015).

Finding 2: Public and private research institutions perform complementary functions that together strengthen the national agricultural innovation system (Fuglie & Heisey, 2007; USDA ERS, 2019; Wang et al., 2013).

Finding 3: Research produces its greatest agricultural benefits when effective systems exist to translate, communicate, adopt, and refine new knowledge and technologies (NIFA, 2025; Wang et al., 2012).

Finding 4: Agricultural innovation is cumulative and adaptive, with researchers, institutions, and producers continually learning from one another and building upon previous advances (Wang et al., 2012; Wang et al., 2015; USDA ERS, 2026).

The historical productivity record provides the first major foundation for this assessment:

American agriculture has increased its output largely by becoming more capable of using its available resources. The next question is what happens when those resources themselves begin to constrain further gains. The following chapter examines natural capital—the soils, water, biodiversity, and ecological processes that provide the biological foundation for agricultural production.

Natural Capital and the Foundations of Agricultural Production

Innovation has allowed American agriculture to produce more from a given resource base, but technology cannot remove the biological foundations of production. Every agricultural system ultimately depends on soils, water, biological diversity, and the ecological processes that support plant and animal growth. These resources are often treated as environmental concerns separate from agricultural productivity, but they are better understood as forms of natural capital: productive assets whose condition directly affects what agriculture can sustain over time (USDA NIFA, 2025; USDA NRCS, 2026).

Natural capital differs from manufactured capital in an important respect. Many ecological processes operate slowly, across large landscapes, and cannot simply be replaced when they are degraded. Soil formation and recovery, groundwater recharge, nutrient cycling, pollination, and biological regulation can depend on processes that develop over long periods. Agricultural

technologies can make more efficient use of these resources, but they cannot fully substitute for the underlying systems themselves (National Academies of Sciences, Engineering, and Medicine [NASEM], 2024; USDA NRCS, 2026).

The historical development of American agriculture also shows that productivity and resource stewardship do not have to be opposing objectives. Conservation practices, improved irrigation, integrated pest management, soil-health management, and other approaches can simultaneously protect natural resources and support agricultural performance. The long-term question is therefore not whether agriculture should choose between production and stewardship, but whether it can maintain the resource base while continuing to improve how that resource base is used (USDA NRCS, 2026; USDA ERS, 2026).

This chapter examines the principal components of agricultural natural capital—soils, water resources, biodiversity, and ecosystem functions—and considers how each affects the long-term productivity and resilience of American agriculture.

Soil Resources

Soil is one of the most fundamental productive assets in agriculture. It provides physical support for plants, stores and supplies water and nutrients, supports extensive biological communities, and mediates processes that influence both crop productivity and environmental quality (USDA NIFA, 2025; USDA NRCS, 2026).

Soils are also dynamic systems rather than fixed production media. Their physical structure, organic matter, nutrient availability, biological activity, water-holding capacity, and infiltration characteristics can change in response to management and environmental conditions. USDA

NRCS identifies soil organic matter, infiltration, structure, water-holding capacity, nutrient availability, microbial activity, and related characteristics as important indicators of soil function and health (USDA NRCS, 2026).

This matters because changes in soil condition can affect both productivity and production costs. Erosion can remove productive topsoil and reduce soil depth. Compaction can interfere with water movement and root development. Nutrient losses can reduce fertility, while declining organic matter can affect soil structure and water retention. These processes may develop gradually, making degradation difficult to recognize through annual production statistics alone (USDA NRCS, 2026; USDA NIFA, 2025).

The history of agricultural conservation demonstrates, however, that soil condition is not simply a fixed constraint. Management practices can improve or protect important soil functions. Reduced tillage, cover crops, diverse rotations, residue management, and other soil-health practices can reduce erosion, improve infiltration, increase organic matter, and support biological activity under appropriate conditions (USDA NRCS, 2026).

The economic implications are significant. USDA NRCS identifies improved soil health as potentially beneficial for productivity and profitability, while conservation practices can reduce erosion, improve nutrient cycling, increase water infiltration, and reduce some input requirements (USDA NRCS, 2026). USDA's conservation programs likewise treat soil health and erosion reduction as agricultural resource concerns rather than purely environmental objectives (USDA ERS, 2026; USDA NRCS, 2026).

Scientific advances are also changing how producers manage soil. Soil testing, digital mapping, remote sensing, field-level indicators, and precision-management technologies allow producers

to distinguish among areas within individual fields and tailor management accordingly. These tools illustrate the relationship between innovation and natural capital: better information can improve resource stewardship, while healthier soils create better conditions for agricultural technologies to perform effectively (USDA NRCS, 2026).

The long-term implication is straightforward. Soil should be treated not merely as an input into agricultural production, but as productive capital that requires continued investment and maintenance. Technology can improve the efficiency with which soil is used, but it cannot eliminate the biological and physical functions that make productive agriculture possible.

Water Resources

Water is another fundamental component of agricultural capacity. Crop production depends on water availability at appropriate times, while livestock systems, processing facilities, and many other agricultural activities also require dependable water supplies. In regions where precipitation is insufficient or unreliable, irrigation and associated infrastructure can substantially expand the range and reliability of agricultural production (Medalie et al., 2025).

The importance of water, however, is not determined simply by annual precipitation. Timing, seasonal variability, drought, groundwater availability, water quality, storage, conveyance, and irrigation efficiency all influence agricultural performance. The USGS finds substantial geographic differences in agricultural water use, with irrigation playing a particularly large role in western regions while other parts of the country rely more heavily on precipitation or have different patterns of water demand (Medalie et al., 2025).

Irrigation illustrates both the productive value of water infrastructure and the scale of agricultural dependence on water resources. USGS estimates indicate that crop irrigation accounted for an average of approximately 105.5 billion gallons per day of withdrawals in the conterminous United States during water years 2010–20, with average consumptive use of approximately 75.7 billion gallons per day. Irrigation accounted for about 43 percent of withdrawals among the three major categories examined in that national assessment—crop irrigation, public supply, and thermoelectric power (Medalie et al., 2025).

Water management has nevertheless become more efficient in important respects. USGS data show that irrigation withdrawals in 2015 were substantially below the historical peak reached in 1980, even though millions of acres continued to be irrigated. The share of irrigated acreage using sprinkler and microirrigation systems also increased between 2010 and 2015, illustrating how changes in technology can affect water-use patterns (Dieter et al., 2018).

These gains do not eliminate the need to maintain the underlying water systems. Groundwater, surface water, reservoirs, and watersheds respond over time to withdrawal patterns, recharge, land management, and climatic conditions. Groundwater in particular can be difficult to manage because aquifer systems differ substantially in their physical characteristics and recharge rates. USGS assessments of groundwater withdrawals demonstrate the importance of aquifers as sources of agricultural and other water supplies across the country (Lovelace et al., 2020).

Water is also subject to competition among users. Agricultural, municipal, industrial, environmental, and other demands may draw upon the same surface-water and groundwater systems. The long-term productive capacity of agriculture therefore depends not only on

improving irrigation efficiency, but also on maintaining water systems capable of supporting agricultural production alongside other legitimate uses (Medalie et al., 2025).

The policy challenge is consequently one of both efficiency and stewardship. Better irrigation technology, monitoring, scheduling, and management can increase the productivity obtained from a unit of water. But efficiency improvements cannot permanently compensate for a water system whose underlying supply is being depleted or whose quality is deteriorating. Agricultural resilience therefore depends upon treating water infrastructure and water resources as complementary components of the same productive system.

Biodiversity and Ecosystem Functions

Agriculture depends on ecological processes that extend beyond the crops and livestock intentionally produced by farmers and ranchers. Pollination, natural pest regulation, nutrient cycling, decomposition, soil biological activity, water filtration, erosion control, and other ecosystem functions can support agricultural production while reducing or moderating the need for some external inputs (USDA Climate Hubs, 2026; USDA Forest Service, n.d.).

These services are often difficult to see in conventional measures of agricultural output because they do not necessarily involve direct market transactions. A population of pollinators does not appear as an agricultural input on a farm balance sheet, and natural pest predators are not purchased in the same way as machinery or fertilizer. Yet ecosystem services can influence production, costs, and resilience by contributing to pollination, pest control, water regulation, soil formation, and other processes that affect agricultural outcomes (USDA Climate Hubs, 2026).

Biodiversity also has value as a reserve of future options. Genetic diversity within crops and livestock provides material that breeders can use to develop varieties and breeds with traits such as disease resistance, drought tolerance, and adaptation to changing conditions. Biological diversity outside agricultural species can likewise support pollinators, natural enemies of pests, and ecological processes that contribute to ecosystem stability (USDA Climate Hubs, 2026; USDA Forest Service, n.d.).

The connection between biodiversity and agricultural resilience becomes especially important when conditions change. A biological trait that provides little commercial value under current conditions may become important when a new pest or disease appears, when climate conditions shift, or when production systems encounter a previously uncommon environmental stress. Maintaining genetic and ecological diversity therefore preserves potential responses that may not be needed immediately but could become valuable in the future.

Agricultural management can also be designed to work with ecosystem functions rather than treating them as separate from production. Integrated pest management, habitat conservation, diversified cropping systems, conservation biological control, and other approaches can maintain ecological functions while supporting agricultural objectives (USDA Climate Hubs, 2026; USDA NRCS, 2026).

This relationship is not without tradeoffs. The National Academies has emphasized that agricultural management can simultaneously increase food production and create pressures on soil health, biodiversity, water quality, and other ecosystem functions. The relevant policy question is therefore not whether every agricultural practice produces exclusively positive or

negative effects, but how management systems can balance multiple functions and sustain the resource base over time (NASEM, 2024).

Biodiversity should consequently be understood as part of agricultural capacity rather than as an external environmental consideration. Healthy ecosystems can support current production, while biological and genetic diversity preserve options for responding to future challenges.

Natural Capital and Innovation: A Reciprocal Relationship

Natural capital and innovation are sometimes treated as separate explanations for agricultural development. In practice, they are closely connected. Innovation allows producers to obtain greater value from land, water, and biological resources, while the condition of those resources determines how effectively new technologies can perform and how long their benefits can be sustained.

The relationship is visible in soil management. Improvements in soil testing, measurement, precision agriculture, and agronomic decision-making can help producers manage resources more precisely, while improvements in soil function can increase water infiltration, nutrient cycling, and other conditions needed for productive agriculture (USDA NRCS, 2026).

The same pattern appears in water management. Irrigation technology and monitoring systems can improve the efficiency of water delivery and application, but the benefits of those technologies ultimately depend on the availability and condition of the water resources themselves (Medalie et al., 2025).

Innovation can therefore stretch the productive potential of natural resources without eliminating their importance. A more efficient irrigation system can reduce water use, but it cannot

indefinitely compensate for a depleted aquifer. Better nutrient management can reduce losses, but it cannot fully replace the biological functions of a healthy soil. Precision agriculture can target inputs more effectively, but it still operates within the biological and physical limits of land and water.

Resource constraints can also encourage innovation. USDA conservation programs increasingly support the development and adoption of tools and practices intended to address soil health, water quality, wildlife habitat, and related resource challenges. Programs such as Conservation Innovation Grants explicitly seek to develop and test new approaches that improve natural resource management on working lands (USDA NRCS, 2026).

This reciprocal relationship has an important implication for how agricultural capacity should be evaluated. Productivity gains are most durable when the resource base supporting them remains functional. Conversely, investments in soil health, water stewardship, biodiversity, and ecosystem function can increase the effectiveness of future technologies by preserving the conditions under which those technologies operate.

Agricultural capacity therefore cannot be understood through a purely technological or purely environmental lens. The historical development of American agriculture has involved both increasing the efficiency with which natural resources are used and developing institutions and practices capable of maintaining those resources.

Synthesis of Natural Capital

The evidence reviewed in this chapter supports a simple but important conclusion: natural capital is productive capital. Soils, water systems, biodiversity, and ecosystem functions are not merely

environmental conditions surrounding agriculture. They are components of the agricultural production system itself. Their condition influences productivity, costs, resilience, and the range of options available to future producers.

The long-term character of natural capital makes stewardship particularly important. Some changes can be managed or reversed, but many ecological processes operate on timescales that are much longer than individual production cycles. Preventing degradation can therefore preserve productive options that might be difficult, costly, or impossible to recover later (NASEM, 2024; USDA NRCS, 2026).

At the same time, stewardship should not be understood as resistance to innovation.

Technological progress can improve resource efficiency, increase the information available to producers, and create new ways of managing environmental constraints. The strongest agricultural systems are those in which innovation and stewardship reinforce one another rather than being treated as competing objectives.

This has a direct implication for the broader assessment. Agricultural capacity is not preserved simply by maintaining current levels of production. It is preserved by maintaining the biological systems that allow future generations to produce, adapt, and innovate. Natural resource stewardship is therefore not separate from long-term productivity policy; it is one of its foundations.

Chapter Findings

Finding 1: Productive soils, reliable water supplies, biodiversity, and ecosystem functions constitute forms of natural capital that directly support agricultural productivity and resilience (USDA NIFA, 2025; USDA NRCS, 2026).

Finding 2: Soil condition is a dynamic determinant of agricultural performance, and management practices that improve soil function can support both productivity and resource resilience (USDA NRCS, 2026).

Finding 3: Water availability, infrastructure, efficiency, and long-term resource conditions jointly determine the reliability of agricultural production, particularly in irrigation-dependent regions (Medalie et al., 2025).

Finding 4: Biodiversity and ecosystem functions provide both present agricultural benefits and future options for responding to biological and environmental change (USDA Climate Hubs, 2026; USDA Forest Service, n.d.).

Finding 5: Innovation and natural resource stewardship are mutually reinforcing. Technologies can improve the efficiency with which natural capital is used, while maintaining that capital increases the effectiveness and durability of technological improvements (USDA NRCS, 2026; Medalie et al., 2025).

The preceding chapters have established two closely related foundations of long-term agricultural capacity: the ability to generate productivity-enhancing innovation and the ability to maintain the natural systems on which production ultimately depends. The next chapter turns to a third foundation—human capital—examining the education, expertise, institutional knowledge,

workforce capacity, and leadership required to connect scientific advances and productive resources to effective agricultural decision-making.

Human Capital and Institutional Knowledge

Agriculture is, at its core, a knowledge-intensive enterprise. Producing crops and livestock requires decisions about genetics, crop and livestock management, irrigation, pest control, soil fertility, machinery, finances, and environmental stewardship, often under conditions that change from one season to the next. As agricultural production has become more technologically sophisticated, the knowledge and skills required to manage it have expanded as well. Human capital—the education, skills, experience, and problem-solving abilities of the people working throughout the agricultural system—is therefore an important component of long-term agricultural capacity (Wang, Hoppe, Hertz, & Xu, 2022).

Human capital extends well beyond farm operators. Researchers, engineers, veterinarians, agronomists, Extension specialists, equipment technicians, managers, and other professionals each contribute different forms of expertise. Their collective knowledge allows scientific discoveries to be translated into practical applications and enables agricultural technologies to be adapted to the wide range of environmental and economic conditions found across the United States (Wang et al., 2022; NIFA, 2026).

The historical productivity record provides evidence that the quality of agricultural labor matters as well as its quantity. USDA Economic Research Service estimates show that the quantity of labor used in U.S. agriculture declined substantially between 1948 and 2019, while labor quality increased. ERS estimates that labor quantity declined by 83 percent over that period, while labor

quality—measured using characteristics including education, age, sex, and employment type—increased by 53 percent. Quality-adjusted labor use nevertheless declined by 74 percent, reflecting the broader shift toward mechanization and other labor-saving technologies (Wang & Njuki, 2025).

These changes illustrate an important distinction. The modernization of agriculture has reduced the need for some forms of manual labor, but it has not reduced the importance of human expertise. Instead, it has changed the kinds of skills agriculture requires. Maintaining agricultural capacity therefore depends increasingly on developing workers who can operate sophisticated equipment, interpret information, manage biological and environmental systems, and make decisions across increasingly interconnected production processes (Wang et al., 2022).

Agricultural knowledge is also cumulative. Each generation inherits scientific research, management practices, institutional experience, and practical lessons developed over decades, but must adapt that knowledge to changing circumstances. New technologies, environmental pressures, market conditions, and emerging pests and diseases continually create new demands for learning. Human capital must therefore be renewed throughout agricultural careers rather than acquired once and treated as permanent (Wang et al., 2022).

Formal education provides one foundation for this process. Universities prepare scientists, engineers, veterinarians, economists, and other professionals whose work advances agricultural research and management. Community colleges, technical schools, vocational programs, and other training institutions provide practical skills for operating and maintaining agricultural equipment and production systems. Continuing education and professional development allow

established workers and producers to update their skills as technologies and production practices change (NIFA, 2025, 2026).

Experience provides another form of knowledge that formal education cannot completely replace. Agricultural production occurs within highly variable biological and environmental systems, and producers develop practical understanding through years of observation, experimentation, and problem-solving. Knowledge about local soils, weather patterns, pests, water availability, crop performance, and operational constraints can be highly valuable even when it is difficult to capture fully in formal publications or training programs. The long-term strength of agricultural human capital therefore depends on combining scientific education with practical experience.

Agricultural Education and Workforce Development

Modern agriculture requires a workforce capable of working with increasingly sophisticated scientific, technological, and managerial systems. The historical shift toward mechanization and productivity-enhancing technology reduced the amount of labor required for many agricultural tasks while increasing the importance of education and technical skills. USDA ERS research finds that rising labor quality, including higher educational attainment, has partially offset the negative contribution of declining labor quantity to agricultural output growth (Wang et al., 2022).

This trend is likely to continue as agricultural systems incorporate increasingly advanced technologies. Precision agriculture, automation, artificial intelligence, biotechnology, remote sensing, robotics, and data-intensive management systems require workers capable of combining agricultural knowledge with expertise in computing, engineering, biology, environmental

science, and other technical fields. USDA NIFA's Education and Workforce Development program explicitly identifies the development of the next generation of research, education, and Extension professionals as a national priority and supports programs intended to address projected shortages of qualified graduates in agriculture, food, forestry, range, and related fields (NIFA, 2026).

Agricultural education should therefore be understood as an investment in productive capacity rather than simply as a response to current labor shortages. Universities, community colleges, vocational programs, Extension services, apprenticeships, internships, and continuing education each provide different pathways into the agricultural workforce and help maintain the technical capabilities required by an increasingly sophisticated sector (NIFA, 2025, 2026).

The diversity of American agriculture makes flexibility particularly important. The knowledge required by a highly mechanized grain operation differs from that required by a cattle ranch, an irrigated specialty-crop operation, or a technologically intensive greenhouse. Workforce development therefore cannot rely entirely on standardized training. It must also provide opportunities for specialization while encouraging interdisciplinary skills that allow professionals to work across traditional occupational boundaries.

Lifelong learning is particularly important because agricultural technologies and operating conditions do not remain static. A worker trained to operate one generation of machinery or information systems may need substantially different skills as those technologies change. Continuing education, professional certification, Extension programming, and employer-sponsored training can help producers and agricultural professionals remain capable of adopting new practices throughout their careers (NIFA, 2025, 2026).

Workforce development must also address demographic change. The hired agricultural workforce is aging, with the average age of foreign-born farmworkers increasing by nearly seven years between 2006 and 2022 as fewer young immigrants entered the sector (USDA ERS, 2025). ERS also identifies aging, migration, and changing workforce composition as important characteristics of contemporary U.S. agricultural labor markets.

These trends make succession and knowledge transfer increasingly important. Retirements among producers, researchers, Extension specialists, technicians, and managers can remove expertise that took decades to develop. At the same time, new workers bring different educational backgrounds and technological skills. Long-term agricultural capacity therefore depends upon maintaining mechanisms through which experience and new knowledge can move between generations.

The increasing complexity of agricultural production also makes interdisciplinary collaboration more important. Research questions and production challenges frequently involve several forms of expertise at once, requiring agricultural professionals to work across biology, engineering, economics, hydrology, computer science, environmental science, and management. Educational systems that prepare professionals to communicate and collaborate across these fields can therefore strengthen the agricultural system's ability to solve problems that do not fit neatly within a single discipline.

Institutional Memory and Intergenerational Knowledge

Not all agricultural knowledge can be found in a textbook, research paper, or technical manual. Farmers and ranchers develop practical understanding through years of experience with particular soils, weather patterns, pests, machinery, markets, and production systems. This

knowledge is especially valuable when decisions depend on local circumstances that are difficult to capture in generalized recommendations.

The transfer of this knowledge occurs through several channels. Family farming operations, mentorships, producer organizations, Extension programs, demonstration farms, professional associations, and collaborative research projects all create opportunities for experienced professionals to share knowledge with those entering the sector. Cooperative Extension, in particular, provides a nationwide educational network connecting research institutions with farmers, ranchers, and rural communities (NIFA, 2025).

Institutional memory extends beyond individual farms. Universities, government agencies, research organizations, and private companies also accumulate knowledge through long-term research programs, datasets, experimental facilities, breeding programs, and organizational experience. Such continuity allows institutions to distinguish temporary fluctuations from longer-term trends and to build new work on the knowledge generated by previous efforts.

Long-term scientific infrastructure provides an especially important form of institutional memory. Research programs that maintain observations across many years can reveal changes that would be difficult to detect through short-term studies. The same principle applies to agricultural institutions more broadly: organizations that preserve data, professional expertise, and historical knowledge give future decision-makers a stronger foundation for responding to new conditions.

The loss of institutional knowledge can therefore create costs that are not immediately apparent. An experienced producer, researcher, Extension specialist, or technical professional may possess practical knowledge that has never been formally recorded. When that individual retires without

an effective mechanism for transferring that knowledge, rebuilding the same expertise may take years—or may not be possible in exactly the same form.

Institutional continuity consequently requires more than simply replacing personnel. Mentorship, succession planning, professional development, archival preservation, and strong professional networks can help ensure that accumulated experience is passed forward rather than lost. These mechanisms also create opportunities for newer professionals to introduce updated scientific and technological knowledge while learning from the experience of those who came before them.

Scientific research and practical experience should therefore be understood as complementary forms of agricultural knowledge. Research provides systematic methods for testing ideas and understanding agricultural systems, while producers provide continual feedback about how those ideas perform under real-world conditions. Maintaining this exchange allows agricultural knowledge to accumulate across generations rather than repeatedly starting over.

Labor, Technical Capacity, and Agricultural Resilience

The agricultural workforce encompasses far more than farmers and ranchers. Seasonal workers, equipment technicians, mechanics, veterinarians, irrigation specialists, engineers, researchers, food safety professionals, logistics personnel, managers, and other workers contribute to the functioning of the agricultural system. The availability and capabilities of this broader workforce therefore influence agricultural capacity as much as the number of farm operators alone (Wang et al., 2022).

Technology has changed the composition of this workforce substantially. Mechanization reduced the demand for many forms of manual labor while increasing the importance of workers able to

operate, maintain, and repair increasingly complex equipment. More recent developments in precision agriculture, remote sensing, geographic information systems, automation, biotechnology, and digital farm management have continued this shift toward specialized technical expertise.

USDA ERS data illustrate the broader transformation. Although agricultural labor input has declined substantially since 1948, the educational and other quality characteristics of agricultural workers have improved. The resulting increase in labor quality has partially offset the effects of declining labor quantity on agricultural output growth (Wang et al., 2022; Wang & Njuki, 2025).

The availability of technical expertise can also influence whether producers can take advantage of new technologies. Sophisticated equipment may require trained technicians for installation and maintenance. Digital management systems may require workers capable of interpreting data. Automated systems may require specialized knowledge to diagnose problems or integrate new tools into existing operations. The value of a technology therefore depends partly on whether the surrounding workforce can support its effective use.

This issue becomes particularly important during periods of disruption. Drought, flooding, disease outbreaks, equipment failures, supply-chain interruptions, and other unexpected events may require rapid decisions under difficult conditions. Technology can expand the available response options, but people still have to recognize the problem, evaluate alternatives, and implement an appropriate response. A workforce with diverse technical and practical skills can therefore strengthen agricultural resilience.

The modernization of agriculture has not made human labor unimportant. It has changed the forms of expertise that agriculture requires. In many parts of the sector, workers increasingly

need to combine knowledge of biology, engineering, technology, economics, and environmental management to address practical production problems.

Maintaining this workforce should consequently be treated as a long-term investment in agricultural capacity. Education, apprenticeships, technical training, professional development, and career pathways can help ensure that producers have access to the expertise needed to operate increasingly sophisticated systems. Strong professional networks can also help maintain access to specialized knowledge when disruptions occur or new technologies emerge.

Leadership, Institutions, and Organizational Capacity

Agriculture does not operate through individual farms and businesses alone. Behind producers is a network of universities, government agencies, Extension organizations, producer associations, cooperatives, conservation districts, private firms, financial institutions, and other organizations that support research, education, resource management, markets, and public administration. The effectiveness of this broader system depends not only on financial resources, but also on institutional continuity, leadership, and the ability to coordinate across organizations.

The importance of organizational capacity is particularly apparent in long-term investments. Research programs, conservation efforts, breeding programs, workforce initiatives, and infrastructure projects often require sustained attention for many years. Effective institutions can preserve long-term priorities even as personnel, administrations, economic conditions, and technologies change.

Cooperative Extension provides a useful example of this institutional continuity. NIFA describes the Cooperative Extension System as a nationwide, non-credit educational network operating

through land-grant universities and local offices, with a presence in or near most of the nation's approximately 3,000 counties. Its role is to connect scientific knowledge with producers, communities, and other stakeholders at the local level (NIFA, 2025).

Institutional coordination is also increasingly important because agricultural challenges rarely fit within a single discipline or organization. A technological innovation may require cooperation among researchers, engineers, private companies, Extension personnel, producers, and government agencies. Effective coordination allows different forms of expertise to be combined and can reduce duplication while making it easier for successful innovations to move into agricultural practice.

Strong institutions are particularly valuable during disruptions. Disease outbreaks, invasive species, extreme weather, natural disasters, supply-chain failures, and other events can affect entire regions or industries at once. Individual producers cannot always address such challenges independently. Institutions capable of gathering information, coordinating technical expertise, mobilizing resources, and communicating effectively can improve the speed and coherence of a collective response.

Institutional strength does not eliminate uncertainty or guarantee successful outcomes. Its value lies in creating the organizational capacity to learn, coordinate, and respond. Strong institutions preserve knowledge, maintain professional relationships, and provide mechanisms through which expertise and resources can be brought together when circumstances change.

Institutional capacity should therefore be understood as another form of productive capital. It complements natural resources, infrastructure, scientific research, and human expertise by providing the organizational framework through which those assets can be used effectively.

Maintaining that capacity will be essential to ensuring that American agriculture can continue to learn, coordinate, innovate, and adapt.

Synthesis of Human Capital

Human capital is the connective tissue of the agricultural system. Scientific discoveries depend on researchers who can expand knowledge, educators who can communicate it, technical specialists who can implement increasingly sophisticated technologies, and producers who can adapt those advances to the realities of their operations. Without these forms of expertise working together, investments in research, infrastructure, and natural resource stewardship cannot reach their full potential.

The historical evidence also indicates that agricultural labor should not be evaluated solely by the number of workers employed. The quantity of agricultural labor has declined substantially over time, while worker characteristics associated with labor quality, including education, have improved. This shift has allowed agricultural output to continue growing as the sector has adopted increasingly labor-saving technologies (Wang et al., 2022; Wang & Njuki, 2025).

Human capital connects the other foundations examined in this assessment. It allows scientific knowledge to move from research institutions into agricultural practice, helps producers manage natural resources, and provides the judgment necessary to respond when technologies or environmental conditions change. In this sense, knowledge is not simply another input. It is part of the mechanism through which other forms of productive capital become useful.

Agricultural expertise is also cumulative. Education provides foundational knowledge, professional experience builds judgment, and institutions preserve lessons that might otherwise

be lost between generations. At the same time, agricultural knowledge must continually evolve as scientific discoveries, technologies, markets, and environmental conditions change.

Maintaining human capital therefore requires both preserving accumulated knowledge and creating opportunities for future generations to expand upon it.

This combination of continuity and renewal is one of the defining characteristics of a resilient agricultural system. The objective of workforce policy should not simply be to replace workers who leave agriculture, but to ensure that each generation enters the sector with enough knowledge and opportunity to build upon what came before.

Chapter Findings

Finding 1: Human capital—including education, technical expertise, and practical experience—is a foundational determinant of long-term agricultural productivity (Wang et al., 2022; Wang & Njuki, 2025).

Finding 2: The agricultural workforce has become smaller in quantitative terms but more highly skilled in important respects, increasing the importance of education and technical expertise as agriculture becomes more technologically sophisticated (Wang et al., 2022; Wang & Njuki, 2025).

Finding 3: Lifelong education, Extension, professional development, and workforce programs are important mechanisms for maintaining and renewing agricultural expertise as technologies and production systems change (NIFA, 2025, 2026).

Finding 4: Intergenerational knowledge transfer and institutional memory help preserve practical expertise while allowing new scientific and technological capabilities to enter the agricultural system.

Finding 5: Strong organizational capacity allows agricultural institutions to coordinate expertise, preserve knowledge, respond to disruptions, and sustain long-term programs even as personnel and external conditions change.

The central lesson of this chapter is straightforward: agricultural capacity ultimately depends on people as much as it depends on land, technology, and infrastructure. Innovation requires skilled researchers and producers. Stewardship requires informed judgment. Resilience requires people and institutions capable of recognizing problems, developing solutions, and adapting when conditions change.

The next chapter turns to another essential foundation of agricultural capacity: physical infrastructure. Roads, railways, ports, energy systems, irrigation networks, storage facilities, communications systems, and processing capacity determine whether agricultural production can move efficiently from farms and ranches into domestic and international markets.

Physical Infrastructure and Agricultural Capacity

Agriculture does not end at the edge of a field or pasture. Modern production depends on an extensive physical network that moves inputs to farms, moves commodities to markets, stores and processes what producers harvest, supplies energy and water, communicates information, and supports scientific research. Roads, railways, inland waterways, ports, irrigation systems, electrical networks, telecommunications, storage facilities, processing plants, and research

infrastructure therefore function as productive assets rather than merely as background services (USDA AMS, 2020, 2026).

The importance of infrastructure depends not only on whether it exists, but on its reliability, capacity, condition, and ability to adapt as agricultural systems change. A transportation bottleneck can raise shipping costs, inadequate storage can limit marketing options, unreliable power can interrupt irrigation or livestock operations, and insufficient broadband can restrict the use of digital agricultural technologies. Because these systems are interconnected, weakness in one part can create consequences elsewhere in the agricultural economy (USDA AMS, 2020, 2026; USDA, 2019).

Infrastructure also expands what producers are able to do. Reliable transportation connects geographically dispersed farms and ranches to suppliers and markets. Irrigation and water-storage systems allow producers to manage variable water availability. Communications infrastructure enables increasingly data-intensive production systems, while storage and processing facilities allow commodities to retain value after leaving the farm. These capabilities reduce constraints imposed by distance, time, and environmental variability and allow agricultural production to function at national and international scales (USDA AMS, 2026; USDA ERS, 2026).

Physical infrastructure has also accumulated over generations. Maintaining it therefore requires continued investment rather than a one-time commitment. Roads, bridges, irrigation networks, water-control systems, electrical infrastructure, storage facilities, and research assets all require maintenance and periodic modernization. As technologies change, infrastructure must also be

adapted to support new methods of production. Infrastructure policy should consequently be understood as an ongoing investment in agricultural capacity.

Transportation Networks

Agricultural production depends on reliable transportation from the moment inputs are purchased until products reach their final markets. Seed, fertilizer, feed, fuel, machinery, and other inputs must reach farms when they are needed, while harvested crops and livestock products must move to storage facilities, processors, distributors, domestic consumers, and export markets (USDA AMS, 2020, 2026).

The U.S. agricultural transportation system is multimodal, combining highways, railroads, inland waterways, ocean shipping, and, for some products, air freight. Different commodities depend on different combinations of these modes. Bulk agricultural commodities often rely heavily on rail and inland waterways, while highways provide essential first- and last-mile connections and remain particularly important for products that require flexible or time-sensitive transportation (USDA AMS, 2020, 2026).

The scale of this network illustrates how closely agricultural competitiveness is tied to transportation infrastructure. USDA reports that agricultural movements account for approximately 30 percent of the ton-miles moved on the U.S. transportation system comprising highways, waterways, and railroads. USDA analysis also found that 80 percent of domestic agricultural commodities travel on roughly 17 percent of U.S. highway mileage, making a relatively small set of high-volume corridors especially important to agricultural freight movement (USDA AMS, 2020, 2026).

Transportation efficiency affects agricultural competitiveness through both cost and reliability. When roads, railways, waterways, and ports function efficiently, producers and businesses can move inputs and commodities at lower cost and respond more readily to market opportunities. Congestion, deterioration, or disruptions can have the opposite effect by increasing operating costs and reducing flexibility (USDA AMS, 2020).

A diversified transportation system also provides an important source of resilience. The ability to shift shipments among highways, rail lines, waterways, and ports can reduce the consequences of disruptions affecting individual routes. Recent USDA analysis of agricultural transportation has emphasized that the major freight modes operate as an interconnected network and that transportation bottlenecks can affect the efficiency and competitiveness of the broader agri-food system (USDA AMS, 2026).

Inland waterways illustrate the importance of this redundancy and specialization. USDA reports that the nation's primary inland waterways system moves approximately 65 percent of U.S. grain exports, providing a comparatively low-cost means of moving bulk commodities from agricultural regions to export markets (USDA AMS, 2026).

Transportation infrastructure therefore does more than move agricultural products from one location to another. It determines how easily producers can obtain inputs, reach markets, respond to changing demand, and participate in domestic and international trade. Maintaining and modernizing transportation networks is consequently an important component of preserving agricultural capacity.

Storage, Processing, and Value-Added Infrastructure

Agricultural production does not end at harvest. Grain elevators, cold-storage facilities, livestock handling systems, packing facilities, food-processing plants, and distribution centers preserve, transform, and distribute agricultural products after they leave the farm. These facilities allow agricultural systems to coordinate production across seasons and regions while reducing the amount of value lost between production and consumption.

Storage provides producers with flexibility in deciding when and where to market their products. Grain and other durable commodities can be held rather than sold immediately after harvest, allowing producers and supply-chain businesses to manage seasonal flows more effectively. For perishable products, refrigerated storage and temperature-controlled transportation are essential to maintaining quality and limiting losses.

Processing infrastructure adds further value by transforming raw agricultural commodities into food, feed, fiber, bio-based materials, and other products. It also connects producers to markets that may not otherwise exist at the local level. The location and capacity of processing facilities can therefore influence regional agricultural development, particularly for commodities that require specialized handling or nearby processing.

This relationship is especially important because gains in farm productivity do not automatically translate into gains for the broader agricultural economy. Larger harvests require adequate storage, transportation, processing, and distribution capacity. Without those complementary systems, constraints downstream can prevent producers from fully realizing the economic value of additional output.

The broader agricultural supply chain therefore needs to be considered as part of productive capacity. Storage and processing infrastructure allow agricultural systems to absorb production,

preserve quality, reduce losses, diversify markets, and move products efficiently toward consumers. Investments in these systems reinforce productivity gains made at the farm level by ensuring that additional production can be handled effectively.

Water Control, Irrigation, and Resource Infrastructure

Water infrastructure determines how effectively agriculture can use the water available to it. Reservoirs, canals, pumping systems, drainage networks, irrigation districts, conveyance systems, and on-farm irrigation equipment allow water to be stored, transported, measured, and applied according to production needs (Hrozencik et al., 2021).

The role of this infrastructure differs substantially across the country. In wetter regions, drainage systems can be essential for removing excess water and maintaining productive fields. In arid and semi-arid regions, irrigation may determine whether intensive agricultural production is feasible in the first place. Agricultural water infrastructure must therefore be designed and managed according to local hydrological and production conditions rather than according to a single national model.

The scale of U.S. irrigation development demonstrates the importance of infrastructure investment. USDA ERS reports that irrigated agricultural land expanded more than tenfold between 1890 and 2022, reaching 54.9 million acres in 2022. The expansion reflected both public and private investment in surface-water infrastructure as well as improvements in groundwater wells and pumping technology (USDA ERS, 2026).

At the same time, agricultural water use has become more efficient. USDA ERS estimates that average water use per irrigated acre declined from more than 2 acre-feet in 1969 to

approximately 1.52 acre-feet in 2023, reflecting the adoption of pressurized irrigation systems and improvements in water application technologies (USDA ERS, 2026).

These improvements demonstrate the importance of combining infrastructure with technology. More efficient irrigation equipment, improved measurement, automated controls, and better management can allow producers to obtain more agricultural output from existing water supplies. Yet efficiency alone cannot eliminate hydrological constraints. Where water supplies are limited, maintaining the infrastructure that stores, conveys, measures, and manages water remains essential.

Infrastructure also provides a buffer against uncertainty. Storage capacity, flexible distribution systems, drainage, and irrigation networks can give producers more options during periods of drought, flooding, or other changes in water availability. These systems cannot eliminate environmental constraints, but they can expand the range of feasible responses.

Maintaining existing infrastructure is therefore as important as constructing new systems. USDA's national Survey of Irrigation Organizations, conducted with USDA NASS and other partners, specifically examines water infrastructure, metering, pricing, governance, and organizational finances, underscoring the institutional and physical dimensions of maintaining reliable agricultural water delivery (Hrozencik et al., 2026).

The long-term objective should consequently be to maintain and modernize water systems while improving the efficiency with which available supplies are used. Agricultural resilience depends not simply on having more water infrastructure, but on having infrastructure capable of operating reliably under changing environmental and production conditions.

Research Infrastructure

Scientific progress depends upon physical and informational infrastructure capable of supporting research over long periods. Experimental farms, long-term field trials, laboratories, breeding facilities, germplasm repositories, environmental monitoring networks, and research databases provide the foundation for generating and preserving agricultural knowledge.

Long-term research is particularly important because many agricultural processes cannot be understood within a single growing season. Soil conditions, crop performance, pest dynamics, water use, ecosystem responses, and the effects of management practices may change slowly or become visible only after years of observation. The USDA Agricultural Research Service's Long-Term Agroecosystem Research (LTAR) Network illustrates the value of this approach. The network brings together long-established research sites across major U.S. agroecosystems and combines coordinated experimentation with long-term measurements to study agricultural productivity, environmental conditions, and emerging technologies (USDA ARS, 2026).

Many LTAR sites had already accumulated decades of research before being incorporated into the network. USDA notes that some locations began experimentation more than a century ago, while the network as a whole includes sites with an average research history of more than 50 years. These long records provide continuity that short-term studies cannot easily reproduce (USDA ARS, 2026).

Research infrastructure also provides places where emerging technologies can be tested before widespread adoption. Coordinated research across different regions allows scientists to determine whether new management practices or technologies work under different environmental and

production conditions rather than assuming that an innovation will perform uniformly everywhere (USDA ARS, 2026).

Biological collections provide another form of research infrastructure. USDA's agricultural germplasm system maintains and distributes genetic resources used in crop research and breeding. These collections preserve biological diversity that can become valuable when researchers seek traits associated with disease resistance, environmental adaptation, or other agricultural characteristics (USDA ARS, 2026).

The value of research infrastructure therefore extends beyond the facilities themselves. A field station, laboratory, dataset, or germplasm repository may support research today while also providing information and biological materials that future researchers will use to investigate problems that cannot yet be fully anticipated. Maintaining these resources allows scientific knowledge to accumulate rather than requiring each generation to rebuild the evidence base from the beginning.

Research infrastructure should consequently be regarded as an investment in future agricultural capacity. Its benefits may be difficult to measure in the short term, but the long-term value lies in preserving the ability of future researchers to understand changing agricultural systems, evaluate emerging technologies, and develop responses to new challenges.

Infrastructure as an Integrated System

Transportation, water, energy, communications, storage, processing, and research infrastructure are often discussed separately because they fall under different institutions and require different forms of expertise. In practice, however, agricultural production depends on all of them

functioning together. Modern agriculture involves continuous flows of materials, energy, information, and products across large geographic areas, so a disruption in one component can create consequences elsewhere in the system.

The interdependence is easy to see. A technology developed through agricultural research is of limited value if producers cannot obtain the necessary equipment or inputs. Irrigation systems depend on energy and water-delivery infrastructure. Precision agriculture requires communications systems capable of transmitting and processing data. Harvested commodities then depend on transportation, storage, processing, and distribution systems before they reach consumers or export markets. These relationships mean that the effectiveness of one infrastructure asset can depend on the condition of others.

Digital agriculture makes this integration even more apparent. USDA's analysis of precision agriculture identifies broadband connectivity as an important enabling condition for technologies involving connected equipment, data collection, and decision support. USDA estimated that increased adoption of digital agriculture technologies, together with expanded rural broadband connectivity, could generate substantial economic benefits, illustrating the degree to which communications infrastructure can influence the value of technological investment (USDA, 2019; McFadden, Njuki, & Griffin, 2023).

The same principle applies to transportation. Highways provide first- and last-mile connections to rail, barge, and ocean transportation, while railroads and inland waterways move large volumes of agricultural commodities over long distances. USDA's agricultural transportation research treats these modes as parts of a connected freight network rather than as wholly separate systems (USDA AMS, 2020, 2026).

The integrated nature of infrastructure also has important implications for resilience. Multiple transportation routes, distributed storage capacity, backup energy systems, and alternative communications pathways can provide options when individual assets fail. Redundancy may appear inefficient when judged only under normal operating conditions, but it can become highly valuable during severe weather, equipment failures, supply-chain disruptions, or other emergencies.

This is particularly relevant as agriculture becomes increasingly dependent on automation and real-time information. As production systems become more interconnected, failures can propagate more quickly across infrastructure networks. Future investments should therefore consider not only whether individual assets are modern and functional, but also whether they are compatible with and resilient within the broader systems on which agricultural production depends.

Agricultural infrastructure should consequently be understood as a system of connected productive assets rather than a collection of independent facilities. Its value comes partly from the relationships among its components: infrastructure moves resources, energy, information, and agricultural products through the system and allows innovations developed in one part of the economy to be used elsewhere.

Synthesis of Physical Infrastructure

Physical infrastructure expands what agriculture can accomplish by reducing constraints imposed by distance, geography, time, and environmental variability. Transportation networks connect producers with suppliers and markets. Irrigation and water-control systems increase production reliability. Storage and processing facilities preserve products and allow producers to capture

value beyond the farm gate. Energy and communications systems support increasingly sophisticated production methods, while research infrastructure provides the physical and informational foundation for continued scientific progress (USDA AMS, 2020, 2026; USDA ERS, 2026; USDA ARS, 2026).

The historical development of American agriculture also shows that infrastructure and innovation have evolved together. Technological changes can create new infrastructure needs, while improvements in infrastructure can accelerate the adoption and usefulness of new technologies. Digital agriculture provides a contemporary example: the potential value of precision technologies depends partly on the availability of reliable broadband and communications infrastructure (USDA, 2019; McFadden et al., 2023).

Infrastructure also contributes to resilience. Reliable transportation routes, flexible water systems, dependable energy, distributed storage and processing, and effective communications provide alternatives when individual parts of the agricultural system are disrupted. The value of that redundancy is particularly apparent in a sector whose production is geographically dispersed and whose supply chains connect farms to distant domestic and international markets (USDA AMS, 2020, 2026).

The cumulative nature of infrastructure creates an important policy consideration. Roads, irrigation systems, water facilities, storage structures, communications networks, and research sites represent investments made over many years. Their value depends not only on construction but also on maintenance, modernization, and the institutional capacity to operate them effectively.

This reflects a broader conclusion that appears throughout the assessment: agricultural capacity does not reside in any single asset. Productive soils, scientific knowledge, skilled people, functioning institutions, physical infrastructure, and economic resources reinforce one another. Investment in one component can be less effective when complementary systems are weak, while coordinated improvements can generate benefits throughout the agricultural economy.

Chapter Findings

Finding 1: Physical infrastructure expands agricultural capacity by reducing geographic, temporal, logistical, and environmental constraints on production and market access (USDA AMS, 2020, 2026).

Finding 2: Transportation, water management, energy, communications, storage, processing, and research infrastructure function as complementary components of an interconnected agricultural system (USDA AMS, 2026; Hrozencik et al., 2026).

Finding 3: Continued maintenance and modernization are essential to preserving infrastructure reliability and allowing agricultural systems to adopt emerging technologies (USDA AMS, 2020; USDA, 2019; USDA ERS, 2026).

Finding 4: Infrastructure investments can generate benefits beyond individual assets because agricultural production depends on connected networks of transportation, water, energy, communications, storage, processing, and research capacity (USDA AMS, 2026; USDA ARS, 2026).

The first four analytical chapters have now examined four major foundations of agricultural capacity: innovation, natural capital, human capital, and physical infrastructure. Together, they

explain much of how American agriculture has been able to increase productivity while adapting to changing conditions.

The next chapter turns to a different but equally important foundation: economic institutions, markets, and financial capacity. Productive resources and infrastructure determine what agriculture is capable of producing, but farms and agricultural businesses must also remain economically viable if those assets are to be maintained, technologies adopted, and productive capacity renewed over time.

Economic Institutions, Markets, and Financial Capacity

Agricultural productivity ultimately has to work in economic terms as well as biological ones. Scientific knowledge, productive soils, skilled workers, and reliable infrastructure can expand what agriculture is capable of producing, but those gains cannot be sustained if agricultural businesses lack the financial capacity to maintain their assets, adopt new technologies, and remain viable over time. Markets, financial institutions, insurance systems, property rights, contracts, and other economic institutions therefore form an important part of the broader system supporting agricultural capacity (USDA ERS, 2025, 2026).

Markets perform several functions within this system. Prices communicate information about supply, demand, scarcity, costs, and expected returns, helping producers and businesses decide how to allocate land, labor, capital, and other resources. Agricultural markets are also unusually exposed to uncertainty because production decisions must often be made months or years before the outcome is known. Weather, disease, prices, input costs, trade conditions, and other factors can all change after resources have already been committed (USDA ERS, 2025).

Financial institutions provide another essential link. Credit allows producers to acquire land, machinery, livestock, irrigation systems, storage, and other long-lived assets whose benefits are realized over time. Financial resilience also gives otherwise viable businesses greater capacity to absorb temporary setbacks without abandoning productive investments. USDA ERS's farm-income and finance program accordingly tracks not only income, but also assets, debt, wealth, solvency, liquidity, and other indicators of financial condition (USDA ERS, 2026).

Risk management connects these economic functions. Producers cannot eliminate uncertainty, but they can use insurance, diversification, contracts, hedging, storage, liquidity, debt management, and other strategies to limit the consequences of individual shocks. Recent ERS analysis shows that U.S. farms use combinations of on-farm, market-based, governmental, and longer-term strategies, with the mix varying according to farm size, commodity specialization, and other characteristics (Baldwin, Miller, & Todd, 2026; USDA ERS, 2025).

The broader implication is that agricultural capacity depends not only on whether agriculture can produce, but also on whether agricultural enterprises have sufficient economic stability to maintain the productive assets inherited from previous generations and invest in the next generation of productivity.

Markets and Resource Allocation

Agriculture continually requires decisions about how to allocate scarce resources among competing uses. Land, water, labor, machinery, energy, and financial capital can all be deployed in different ways, and producers must continually assess where those resources are likely to produce the greatest return. Markets provide one of the principal mechanisms through which these decisions are coordinated.

Prices are an important part of that process. A producer deciding what to plant, whether to expand a livestock operation, or whether to purchase new machinery considers expected costs and returns. Input suppliers respond to demand for seed, fertilizer, equipment, and other products. Processors and distributors adjust their operations as consumer preferences and market conditions change. Millions of individual decisions are therefore connected through market signals, allowing resources to move toward uses that appear more productive under prevailing conditions (USDA ERS, 2025).

Agricultural markets cannot respond instantly to those signals, however. Crops take time to grow, livestock require extended production cycles, and major investments in land and equipment are often difficult to reverse. Weather, disease, and other environmental conditions can also change the outcome after production decisions have already been made. These characteristics contribute to short-term volatility while still allowing markets to influence resource allocation over longer periods (USDA ERS, 2025).

Markets also depend upon institutions that make exchange possible. Contracts establish the terms of transactions, financial institutions provide credit, transportation systems connect buyers and sellers, and information systems allow producers and businesses to make decisions with greater knowledge of prices and conditions. Property rights and legal institutions likewise provide greater certainty over the use of productive assets and the enforcement of commercial agreements.

These supporting institutions can be easy to overlook because they tend to become most visible when they fail. A functioning agricultural market depends upon much more than the existence of

buyers and sellers. It requires systems that allow participants to identify opportunities, obtain financing, verify quality, move products, manage risk, and enforce agreements.

Markets should therefore be understood as coordination systems rather than simply places where commodities are bought and sold. Their importance to agricultural capacity lies in their ability to guide resource allocation while providing incentives for producers and businesses to respond to changes in demand, technology, environmental conditions, and production costs.

Capital Investment and Access to Finance

Agriculture requires substantial amounts of capital. Productive land, machinery, irrigation systems, livestock facilities, storage capacity, processing equipment, and digital technologies can all require large investments before they begin generating returns. Maintaining agricultural capacity therefore depends not only on the existence of these assets, but also on whether producers can finance their acquisition, maintenance, replacement, and modernization.

Credit is particularly important because many agricultural investments have long useful lives and require substantial upfront expenditures. Producers may need to finance equipment, land, livestock, irrigation systems, or other improvements while returns are realized gradually over multiple production cycles. Access to financing allows these costs to be distributed over the period during which the investment generates value rather than requiring every major investment to be funded from current operating income.

The importance of financial capacity is reflected in the way USDA ERS evaluates farm-sector performance. Its financial measures include farm receipts and expenses, net farm income, assets, debt, wealth, and financial ratios associated with solvency, liquidity, and efficiency. Together,

these measures provide a broader picture of whether the agricultural sector can maintain and renew its productive assets (USDA ERS, 2026).

Investment decisions are necessarily made under uncertainty. Producers must consider expected commodity prices, input costs, interest rates, weather, and other risks when deciding whether a major investment is economically justified. An investment that appears attractive under one set of conditions may become much less so under another. Financial resilience therefore includes not only access to credit but also sufficient liquidity and flexibility to withstand periods of weaker income.

Access to capital can also affect the pace at which new technologies spread. An innovation may require complementary investments in buildings, specialized equipment, data systems, maintenance, or workforce training. A producer may therefore face a substantial capital requirement even when the underlying technology is expected to improve productivity or reduce costs. Limited or expensive credit can delay modernization even when a technology is technically promising.

The importance of financing consequently extends beyond the individual operation. When producers can replace aging equipment, modernize facilities, and adopt productive technologies, the broader agricultural system is better positioned to preserve productivity over time. Conversely, prolonged financial stress can delay reinvestment and gradually weaken the productive assets on which future output depends.

Current conditions illustrate why financial resilience matters. USDA ERS's 2026 projections indicate that farm-sector income remains substantial but continues to vary across commodities, regions, and farm types, while the Federal Reserve Bank of Kansas City reported that farm

financial conditions had tightened gradually amid weak crop profitability and elevated production costs (USDA ERS, 2026; Federal Reserve Bank of Kansas City, 2026).

Long-term agricultural policy should therefore distinguish temporary financial difficulty from deeper deterioration in productive capacity. A sector can experience a weak year without losing its underlying productive strength, but prolonged inability to maintain and replace productive assets can gradually erode that strength.

Risk Management and Agricultural Stability

Uncertainty is an unavoidable part of agricultural production. USDA ERS identifies five broad categories of farm risk: production risk, price or market risk, financial risk, institutional risk, and human or personal risk. Weather, disease, pests, changing prices, input costs, government policies, debt obligations, and unexpected personal events can all affect the financial performance of agricultural enterprises (USDA ERS, 2025).

Risk management does not eliminate these uncertainties. Its purpose is to prevent an individual shock from producing consequences that threaten the longer-term viability of an otherwise productive enterprise. Producers use a range of strategies for this purpose, including enterprise diversification, insurance, production and marketing contracts, hedging, storage, liquidity management, debt management, and off-farm income (USDA ERS, 2025).

The appropriate combination varies considerably among farms. A diversified operation may use different crops or livestock enterprises to reduce dependence on a single source of income. A producer with substantial exposure to price movements may use forward contracts or futures and options. Another operation may rely more heavily on insurance, savings, or liquidity. ERS

research finds that the use of these strategies differs with farm size and commodity specialization, reinforcing the importance of flexible approaches rather than a single national model of risk management (Baldwin et al., 2026).

Risk management also affects the willingness to invest. Adopting a new technology or management practice involves uncertainty, particularly when the initial cost is substantial or the expected benefits have not been demonstrated under local conditions. Producers with greater financial and operational flexibility may be better positioned to absorb a disappointing outcome and therefore more willing to experiment. Conversely, persistent financial stress can encourage producers to postpone investments even when they might increase long-term productivity.

This relationship makes risk management part of the innovation system as well as the financial system. New technologies can reduce certain production risks, while effective risk-management tools can make it easier for producers to undertake investments whose benefits are uncertain. Risk management therefore helps create the conditions under which continual experimentation and improvement can occur.

Agricultural resilience should consequently not be understood as the absence of volatility. Production and prices will continue to fluctuate. A resilient agricultural enterprise is one that can experience a bad season without losing the ability to maintain its productive assets, continue operating, and invest again when conditions improve.

This distinction is important at the national level as well. When individual agricultural businesses can absorb temporary shocks, disruptions are less likely to become permanent losses of productive capacity. The ability of the sector to recover from difficult periods therefore depends partly on the financial resilience of the enterprises that make up the agricultural economy.

Competition, Innovation, and Market Dynamism

Agricultural productivity depends not only on having functioning markets, but also on whether those markets encourage experimentation, efficiency, and continued improvement. Competitive pressures can encourage producers and businesses to reduce costs, improve products, develop new technologies, and respond to changing consumer preferences.

Competition also creates incentives to experiment. Producers compare production systems, technologies, machinery, crop varieties, and management practices, while input suppliers, equipment manufacturers, processors, and agricultural technology firms compete by offering products and services that provide greater value. The resulting process of adoption, evaluation, refinement, and replacement can contribute to productivity growth over time.

Markets also facilitate structural adaptation. Consumer preferences change, export opportunities open and close, technologies evolve, and production conditions vary. Producers respond by changing what they produce, how they produce it, how their businesses are organized, and where they invest. The cumulative effect of these adjustments can gradually change the structure of agricultural production.

The effectiveness of competition depends, however, on the institutions surrounding it.

Participants need reliable market information, enforceable contracts, secure property rights, and predictable commercial relationships. Without these foundations, transaction costs and uncertainty can interfere with otherwise productive investment decisions.

Competition should therefore be evaluated as part of a larger institutional environment rather than treated as an end in itself. A dynamic agricultural economy needs markets that allow

resources to move toward productive opportunities while maintaining the institutional conditions that permit businesses and producers to make informed long-term decisions.

The broader principle is consistent with the productivity evidence examined earlier in this assessment: agricultural capacity cannot remain static. Technologies change, resource availability shifts, consumer preferences evolve, and new risks emerge. An economic system that allows producers and businesses to adjust to those changes is better positioned to preserve productivity over time.

Economic Resilience and Long-Term Capacity

Short-term agricultural performance and long-term agricultural capacity are related, but they are not the same thing. A year of high commodity prices may increase farm income without necessarily strengthening the underlying productive base. Conversely, a difficult year may reduce income without permanently damaging an otherwise healthy agricultural operation.

Long-term capacity depends more fundamentally on whether producers can continue maintaining land and equipment, adopting technologies, investing in infrastructure and human capital, and responding to new opportunities. Economic resilience is the ability to withstand temporary setbacks without losing those capabilities.

Financial reserves, liquidity, access to credit, manageable debt, diversified income, risk-management tools, and functioning markets can all contribute to that resilience. ERS identifies savings, off-farm employment, insurance, diversification, contracting, and other mechanisms among the strategies used by farms to manage risk (Baldwin et al., 2026; USDA ERS, 2025).

The condition of individual agricultural enterprises also matters for the broader system. Financially stable producers are better positioned to maintain equipment, retain workers, purchase inputs, continue adopting technologies, and recover after disruptions. When large numbers of otherwise viable operations face prolonged financial stress, the resulting reduction in investment can eventually affect local supply chains, rural economies, and the productive capacity of the agricultural sector more broadly.

Recent financial data illustrate why this distinction matters. USDA ERS's current farm-sector reporting tracks changes in assets, debt, wealth, income, liquidity, and solvency rather than relying on production or revenue alone. The Federal Reserve likewise monitors agricultural credit conditions, loan performance, farmland values, and leverage when assessing the health of the agricultural economy (USDA ERS, 2026; Federal Reserve Bank of Kansas City, 2026).

American agriculture has repeatedly adjusted to changing economic conditions. Producers have altered production systems, adopted technologies, reorganized businesses, entered different markets, and changed their approaches to risk. These adjustments have not eliminated economic cycles, but they have helped agricultural enterprises recover and continue investing through periods of uncertainty.

Economic resilience should therefore be understood as the ability to absorb disruption without permanently losing productive capability. A system can experience substantial short-term losses and still remain fundamentally resilient if its productive assets, knowledge, workforce, and institutions remain intact. Conversely, prolonged financial stress can gradually weaken those foundations even when production remains strong in the short term.

Synthesis of Economic Institutions and Market Capacity

Economic institutions provide the framework within which agricultural production is financed, coordinated, and sustained. Markets help allocate resources, financial systems provide capital, risk-management tools help producers cope with uncertainty, and commercial institutions allow agricultural products and inputs to move through increasingly complex supply chains (USDA ERS, 2025, 2026).

This chapter reinforces a broader conclusion that runs throughout the assessment: agricultural capacity is cumulative. Productive land must remain economically worth maintaining.

Infrastructure must be repaired and eventually replaced. Scientific research requires continuing investment. Producers and workers need opportunities to develop and apply their expertise.

Economic institutions influence each of these processes by shaping incentives, access to capital, and the consequences of uncertainty.

Resilient agricultural economies must also balance efficiency with adaptability. Efficient systems use available resources effectively under prevailing conditions. Adaptable systems retain the flexibility to change when those conditions no longer apply. The two qualities are related, but they are not identical. A production system optimized entirely for current prices, technologies, or environmental conditions may perform extremely well in the short run while remaining vulnerable to changes that were not anticipated.

For the same reason, agricultural performance should be evaluated over sufficiently long time horizons. Commodity prices, production volumes, and farm income can change significantly from year to year without fundamentally altering the productive capacity of the sector. More consequential are the conditions that determine whether producers can continue investing in land, equipment, technologies, people, and resource stewardship.

Markets should therefore be understood as part of a larger agricultural system rather than as an independent source of capacity. They influence how resources are allocated and how strongly producers are rewarded for investment and innovation, but they cannot indefinitely compensate for degraded natural resources, inadequate infrastructure, declining expertise, or weak research institutions.

The long-term productive capacity of American agriculture ultimately emerges from the interaction of these different forms of capital. Economic institutions help coordinate and finance them, but they cannot substitute for them. Their importance lies in creating the conditions under which investments in natural resources, knowledge, human expertise, technology, and infrastructure can be maintained and renewed.

Chapter Findings

Finding 1: Market institutions support agricultural capacity by helping allocate productive resources and enabling producers and businesses to respond to changing economic conditions (USDA ERS, 2025).

Finding 2: Access to capital and sustained investment are essential for maintaining productive assets, adopting technologies, and modernizing agricultural operations (USDA ERS, 2026).

Finding 3: Effective risk management helps producers absorb environmental and economic shocks without unnecessarily sacrificing long-term productive capacity (USDA ERS, 2025; Baldwin et al., 2026).

Finding 4: Long-term agricultural performance depends on economic institutions that support investment, innovation, adaptability, and the continued financial viability of agricultural enterprises (USDA ERS, 2025, 2026).

The assessment has now examined five major foundations of agricultural capacity: innovation, natural capital, human capital, physical infrastructure, and economic institutions. None operates independently. Scientific knowledge depends on skilled people and research infrastructure; technology depends on productive natural resources; infrastructure requires continuing investment; and all of these assets ultimately depend on agricultural enterprises being economically capable of maintaining and renewing them.

The next chapter turns to another factor that determines whether these foundations can be carried forward: demographic change. Agricultural capacity ultimately depends on people—not only producers working farms and ranches, but also researchers, technicians, engineers, educators, managers, and institutional leaders. Changes in the age, size, skills, and geographic distribution of this workforce will therefore have important implications for the future of American agriculture.

Demographic Change and Agricultural Continuity

Agriculture depends on more than land, capital, and technology. It also depends on people who can manage those resources, maintain institutions, operate increasingly sophisticated systems, and carry accumulated knowledge into the future. Each generation inherits an agricultural system shaped by the investments and experience of those that came before it. Long-term capacity

therefore depends on whether new participants can enter the sector, develop expertise, assume responsibility for productive assets, and build upon what previous generations have learned.

Demographic change affects agriculture through several interconnected channels. Population growth influences demand for agricultural products, migration affects labor availability and the distribution of population, education shapes the supply of skilled workers, and age structure influences retirement and succession. These processes tend to unfold gradually, but their cumulative effects can become significant over decades.

The demographic environment is also changing in ways that are particularly relevant to rural America. USDA Economic Research Service data show that the nonmetropolitan population has begun growing again after a decade of overall population loss, reaching approximately 46.2 million people in July 2024. That growth has been driven primarily by net migration, which has offset continued natural population decline. At the same time, rural populations are older on average than metropolitan populations, and the working-age rural population has declined since 2010 (USDA ERS, 2025, 2026).

These trends make demographic continuity an important part of agricultural capacity. The challenge is not simply whether there will be enough people working directly on farms. It is whether agricultural regions will continue to have access to the producers, technicians, researchers, educators, businesses, and institutions needed to keep the broader agricultural system functioning.

Generational Succession

Agriculture is unusual in the extent to which its productive assets can remain in use for decades or even generations. Farmland, orchards, vineyards, irrigation systems, livestock facilities, and established farm businesses often outlast the careers of the people who originally developed them. Succession is therefore more than a change in ownership. It is one of the mechanisms through which productive assets, management responsibility, and accumulated knowledge are carried forward.

The age structure of agricultural operators makes succession an increasingly important consideration. USDA ERS notes that many U.S. farmers are relatively advanced in age and that farmland ownership and tenure have important implications for access to land, succession, and opportunities for new and beginning farmers (USDA ERS, 2026). The 2022 Census of Agriculture found that just over 60 percent of U.S. farmland was owner-operated, illustrating the continuing importance of ownership and tenure arrangements to decisions about how agricultural land is transferred and managed.

A successful transition preserves useful knowledge while creating room for change. New producers inherit established production systems, local knowledge, business relationships, and physical assets, but they also bring different educational backgrounds and familiarity with newer technologies and management practices. When these forms of experience are combined effectively, agricultural enterprises can maintain continuity without becoming static.

Succession also matters beyond individual farms and ranches. Research institutions, Extension organizations, conservation districts, agribusinesses, cooperatives, and government agencies all depend upon the transfer of leadership and technical expertise. Experienced professionals eventually retire or move into new roles, and the knowledge associated with their positions may

take years to develop. Maintaining institutional performance therefore requires more than filling vacancies; it requires opportunities for younger professionals to develop expertise and gradually assume greater responsibility.

The economic viability of succession matters as well. Recent USDA ERS research using the 2022 Census of Agriculture found that beginning-farmer operations can face distinctive challenges to business survival and that land tenure, differentiated markets, crop insurance, and government payments were among the characteristics associated with survival over the 2012–2022 period (Lacy et al., 2026).

These findings reinforce a broader point: generational renewal depends upon access to productive resources as well as the willingness of experienced operators to transfer responsibility. A younger producer may have the skills and motivation to enter agriculture but still face substantial barriers if farmland, financing, infrastructure, or markets are inaccessible.

Successful succession should therefore be viewed as a process of renewal rather than simple replacement. The objective is not merely to preserve existing agricultural enterprises in their current form, but to ensure that future producers inherit enough land, knowledge, capital, and institutional support to improve upon what they receive.

Labor Availability and Workforce Composition

Agriculture depends on a workforce that extends far beyond farmers and ranchers. Equipment operators, mechanics, veterinarians, agronomists, food scientists, engineers, logistics specialists, seasonal workers, researchers, managers, and other professionals all contribute to the functioning of modern agricultural systems.

The broader labor market has changed considerably as agricultural production has become more productive and technologically intensive. USDA ERS research shows that agricultural employment and labor inputs have declined substantially over the long term while labor quality has increased. This reflects the shift away from labor-intensive production toward systems that depend more heavily on mechanization, technology, and specialized expertise (Wang et al., 2022; Wang & Njuki, 2025).

The result is an agricultural workforce that is smaller in some respects but more specialized in others. Modern production systems require people who can operate sophisticated machinery, interpret data, maintain equipment, manage biological systems, and integrate technologies into existing operations. This means that labor availability cannot be evaluated solely by counting workers. The capabilities those workers possess matter as well.

Labor needs also differ substantially across regions and production systems. Highly mechanized grain operations have different labor requirements from specialty-crop farms, livestock operations, greenhouses, or other controlled-environment systems. Seasonal planting, harvesting, and processing create additional fluctuations in labor demand. The diversity of American agriculture therefore argues for a flexible workforce rather than a single national model of agricultural employment.

The availability of technically qualified workers can also influence technology adoption. A producer may be able to purchase sophisticated machinery or digital management systems but still require technicians, operators, data specialists, or other professionals to use those tools effectively. The productivity gains associated with technological innovation therefore depend

partly on whether the surrounding labor market can provide the expertise necessary to implement and maintain it.

Labor availability is also influenced by broader demographic trends. USDA ERS reports that rural working-age populations have declined while rural populations have become older. In 2023, people age 65 and older represented 21 percent of the nonmetro population, compared with 17 percent of the metro population; the number of working-age people in rural areas has also declined since 2010 (USDA ERS, 2026).

These trends do not mean that agricultural capacity must inevitably decline. The historical record demonstrates that agricultural productivity can increase while the number of people directly engaged in production falls. What matters is whether technology, education, workforce development, and institutional adaptation can compensate for reductions in labor quantity while maintaining the skills needed throughout the agricultural system.

The challenge is therefore to maintain both sufficient labor and sufficient capability. An agricultural workforce capable of learning, adapting, and working with new technologies contributes not only to current production but also to the sector's ability to incorporate future scientific and technological advances.

Population Change and Rural Communities

Agriculture is closely connected to the economic and institutional health of the communities in which it operates. Farms and ranches rely on surrounding businesses and institutions for equipment repair, financial services, healthcare, education, transportation, supplies, professional

expertise, and other services. These institutions, in turn, depend upon sufficient population and economic activity to remain viable.

Rural demographic conditions vary considerably across the United States. USDA ERS reports that the rural population increased between 2020 and 2024, but growth was uneven. From 2020 to 2024, more than half of nonmetropolitan counties experienced population decline, while growth was concentrated particularly in counties adjacent to metropolitan areas and counties with recreation-oriented economies (USDA ERS, 2025).

This variation matters because population change is not simply a question of whether rural America as a whole is growing. Individual agricultural regions may experience very different demographic pressures. Some communities may attract new residents and maintain or expand services, while others may struggle with declining working-age populations and difficulty sustaining institutions.

Age structure compounds these differences. USDA's 2024 *Rural America at a Glance* found that rural populations were older on average than urban populations and that the population age 65 and older had grown particularly rapidly in rural counties. Age structure affects both labor supply and the demand for services such as healthcare and eldercare, increasing pressure on communities with relatively small working-age populations (Farrigan et al., 2024).

These demographic changes can influence agriculture indirectly. When communities lose residents, schools, healthcare providers, businesses, and other services may become more difficult to sustain. Rural ERS research emphasizes that demographic and economic conditions are closely connected and that educational attainment, employment, and the availability of local services influence rural well-being (USDA ERS, 2025, 2026).

This does not mean that every rural population decline threatens agricultural production. Highly mechanized operations can remain productive with relatively few workers, and agriculture can continue to benefit from services located outside the immediate community. The more important concern is whether the regional network of institutions supporting agricultural production remains sufficiently strong.

Rural communities therefore function as part of the broader infrastructure of agriculture. Schools and colleges help develop future workers, businesses provide equipment and services, healthcare systems support the workforce, and local organizations provide networks through which producers and institutions coordinate. Their condition can influence the costs of doing business and the ability of agricultural enterprises to attract and retain skilled workers.

The relationship is reciprocal. Agricultural activity provides income, employment, markets, and tax revenues that support rural communities, while healthy communities provide services and institutions that make agricultural production more viable. Agricultural policy should therefore recognize that the long-term capacity of farms and ranches is partly dependent on the condition of the communities surrounding them

Demographic Adaptation and Agricultural Innovation

Demographic change affects agriculture not only through the number of people available to work, but also through the knowledge and perspectives entering the agricultural system. Each generation brings different educational backgrounds, technical skills, and familiarity with emerging technologies. Experienced producers, meanwhile, bring practical knowledge developed through years of observation and adaptation to local conditions.

The history of agricultural development illustrates this relationship. Mechanization reduced the need for manual labor while increasing the importance of mechanical and technical skills.

Advances in genetics, chemistry, and agronomy expanded the role of scientific knowledge in production. More recent advances in digital agriculture, automation, remote sensing, and data analysis have created new demands for technical expertise. The transition has therefore involved repeated changes in the capabilities agriculture requires rather than simply changes in workforce size.

Education and knowledge-transfer institutions are important to this process. Universities, Extension services, producer organizations, technical advisers, equipment manufacturers, and professional associations provide mechanisms through which new information reaches producers and practical experience feeds back into technological development. The Cooperative Extension System, for example, maintains a nationwide network designed to connect research-based knowledge with producers and rural communities (NIFA, 2025).

Generational renewal can also expand the range of perspectives available within agriculture. New participants may bring alternative business models, production methods, technologies, or marketing strategies. Experienced producers contribute knowledge of local conditions, historical experience, and practical judgment. Neither perspective is sufficient on its own. Their interaction creates a balance between continuity and experimentation.

Demographic change therefore does not inherently reduce agricultural capacity. Its effects depend in part on whether agricultural institutions can facilitate the exchange of knowledge between generations and provide opportunities for new participants to enter the sector. Where

those mechanisms function well, generational renewal can reinforce innovation rather than disrupt continuity.

The key challenge is consequently not simply replacing an aging workforce. It is ensuring that accumulated knowledge is preserved while new generations acquire the skills and opportunities necessary to build upon it.

Synthesis of Demographic Change

Demographic change is a permanent feature of agricultural development rather than a temporary problem that can simply be solved. Population growth, migration, educational attainment, labor-force participation, aging, retirement, and generational succession have continually reshaped American agriculture. The important question is how those changes affect the agricultural system's ability to renew itself.

The current demographic picture reinforces the importance of that question. Rural America has experienced renewed population growth since 2020, but that growth is uneven, rural populations remain older than metropolitan populations, and working-age populations have declined in many rural areas (USDA ERS, 2025, 2026).

At the same time, agricultural entry and succession remain active processes. USDA ERS's recent analysis of beginning farmers and ranchers demonstrates that new operators continue to enter the sector, while also showing that business survival depends on a combination of land tenure, markets, insurance, government programs, and other characteristics (Lacy et al., 2026).

A central lesson of this chapter is that continuity does not mean maintaining agriculture exactly as it exists today. Productive agricultural systems remain viable because they are capable of

renewal. Technologies change, production systems evolve, institutions adapt, and successive generations take responsibility for managing productive assets.

Demographic continuity is therefore connected to every other component of agricultural capacity examined in this assessment. Scientific research requires trained researchers. Infrastructure requires people capable of operating and maintaining it. Economic institutions require producers and businesses capable of making long-term decisions. Natural resources require knowledgeable stewardship. The continuity of all these systems ultimately depends on people who can assume responsibility for them and adapt them to changing circumstances.

Demographic trends should consequently be evaluated not simply by changes in population size or agricultural employment, but by what those changes mean for the system's capacity to renew itself. The central question is whether future generations will have access to the knowledge, resources, institutions, and opportunities needed to sustain and improve agricultural productivity.

Chapter Findings

Finding 1: Long-term agricultural capacity depends upon successful generational succession that transfers productive assets, knowledge, and management responsibility across generations (USDA ERS, 2026; Lacy et al., 2026).

Finding 2: Workforce capability increasingly depends upon technical, scientific, and managerial expertise as agricultural systems become more technologically sophisticated (Wang et al., 2022; NIFA, 2026).

Finding 3: Rural demographic conditions influence agricultural capacity not only through labor availability but also through the ability of communities to maintain the educational, commercial,

healthcare, financial, and technical institutions that support agricultural enterprises (USDA ERS, 2025, 2026).

Finding 4: Demographic renewal is most likely to strengthen agricultural resilience when institutions facilitate education, mentorship, succession, knowledge transfer, and opportunities for new participants to enter agriculture.

The preceding chapters have examined agricultural capacity across resources, institutions, infrastructure, markets, and people. Together, they describe the structural foundations that have allowed American agriculture to sustain and improve productivity over many decades.

The next chapter turns outward, examining the forces that may alter the environment in which those foundations operate. Environmental change, technological acceleration, global economic and geopolitical developments, and competition for land, water, labor, and capital will all shape the conditions under which American agriculture operates over the coming decades.

Long-Term Drivers of Agricultural Change

American agriculture does not operate under fixed conditions. Climate and environmental conditions change, technologies develop, demographics shift, markets evolve, international trade changes, and geopolitical events alter the economic environment in which producers operate.

These forces differ in their timing and magnitude, but each can affect agricultural production and investment over periods measured in decades.

The distinction between **foundations** and **drivers** is useful here. The preceding chapters examined the capabilities that allow agriculture to remain productive: innovation, natural capital, human expertise, infrastructure, and economic institutions. This chapter examines forces that

change the conditions under which those capabilities must operate. Environmental change may alter production conditions; technological acceleration may change what producers are capable of doing; globalization can expand markets while increasing exposure to external disruptions; and competition for land, water, labor, and capital can change the resources available to agriculture.

These forces rarely act independently. Climate conditions can influence technological priorities and water demand. Demographic change can affect both labor availability and technology adoption. Global supply chains can influence the availability and price of agricultural inputs. Resource competition can encourage innovation while simultaneously increasing pressure on productive land and water. The future agricultural environment will therefore be shaped less by any single trend than by the interaction among them.

Because these changes cannot be predicted precisely over a 25- to 50-year horizon, the objective of this chapter is not to construct a single forecast. Instead, it examines the mechanisms through which major long-term drivers may affect agricultural capacity and the characteristics that allow agricultural systems to adjust as circumstances change.

Environmental Change

Agriculture has always been closely tied to environmental conditions. Temperature, precipitation, water availability, growing-season length, pest pressures, and other ecological conditions influence the biological processes on which crop and livestock production depend. USDA's Climate Hubs report that changes in temperature, precipitation patterns, growing seasons, and extreme precipitation are already affecting agricultural systems across the United

States, although the effects differ substantially by region and commodity (USDA Climate Hubs, 2026).

Long-term environmental change differs from ordinary year-to-year variability mainly in its persistence and cumulative effects. Changes in temperature regimes, precipitation patterns, water availability, and extreme events can gradually alter the conditions under which particular crops and livestock systems are most productive. USDA notes that changing climate conditions can affect agriculture directly through temperature and precipitation and indirectly through changes in pests, pollination, and other ecosystem services (USDA Climate Hubs, 2026).

These effects will not be uniform across the country. USDA's analysis emphasizes that climate impacts vary among regions, crops, and production systems. Some areas may experience longer growing seasons, while others may face greater heat stress, changing precipitation patterns, or increased water constraints. The resulting effects will depend partly on local resources, management practices, technologies, and the capacity of producers to adapt (USDA Climate Hubs, 2026).

Environmental change should therefore be understood not simply as a constraint, but as a continuing driver of agricultural adaptation. Producers have historically responded to environmental conditions by changing crop varieties, planting schedules, irrigation and drainage systems, soil-management practices, and other aspects of production. The ability to make such adjustments is likely to remain important regardless of the precise environmental conditions that emerge.

The capacity to respond depends on the foundations examined earlier in this assessment.

Scientific research expands the range of technologies and management strategies available to

producers. Natural resource stewardship helps preserve the condition of soils, water systems, and ecosystems. Infrastructure provides additional flexibility, while economic institutions provide the capital and incentives required to make adjustments. Human expertise allows producers and institutions to interpret changing conditions and decide which responses are most appropriate.

This relationship means that uncertainty about future environmental conditions does not eliminate the possibility of effective planning. Instead, it increases the value of investments that perform well under multiple conditions. Practices that improve soil health, increase water-use efficiency, strengthen monitoring, diversify production systems, and improve decision-making can provide benefits across a range of possible environmental futures rather than depending on one particular forecast (USDA Climate Hubs, 2026; USDA NRCS, 2026).

The appropriate response to environmental uncertainty is consequently not to attempt to predict every future condition with precision. It is to maintain an agricultural system capable of learning and adjusting as conditions change.

Technological Acceleration

Technological change is likely to remain one of the most important forces shaping American agriculture. Precision agriculture, automation, robotics, biotechnology, artificial intelligence, remote sensing, digital communications, and increasingly sophisticated data systems are expanding the range of tools available to producers and agricultural businesses (McFadden, Njuki, & Griffin, 2023; Wang et al., 2024).

The adoption of digital agricultural technologies has already expanded substantially in several major U.S. crops. USDA ERS reports that automated guidance systems are now used on more

than half of U.S. acreage planted to several major crops, including corn, cotton, rice, sorghum, soybeans, and winter wheat. Other technologies, including yield maps, soil maps, and variable-rate technologies, have also expanded, although their adoption remains much lower for some crops and production systems (McFadden et al., 2023).

These technologies can affect agricultural productivity through several pathways. Sensors and data systems can provide more detailed information about crops, soils, water, and equipment. Automation can reduce the labor required for repetitive tasks. Precision technologies can allow inputs to be applied more selectively, while digital decision-support systems can make information available more quickly to producers. USDA ERS identifies labor-saving benefits, expected productivity improvements, soil variability, technology prices, USDA programs, and access to technical services among the factors influencing adoption (McFadden et al., 2023).

Technology, however, does not generate productivity automatically. Digital agriculture depends on broadband, reliable electricity, equipment maintenance, data systems, technical expertise, and other supporting infrastructure. Producers must also have sufficient information and financial capacity to evaluate and adopt new technologies. Adoption rates vary substantially across crops and technologies, demonstrating that availability alone does not guarantee widespread use (McFadden et al., 2023).

The rapid pace of technological development can also create a different kind of challenge: the need to continually evaluate whether new technologies provide enough value to justify their costs. Producers must consider not only whether a technology works technically, but whether it is compatible with their production system, labor force, financial position, and management capabilities.

This makes technological acceleration another reason to emphasize human capital and institutional capacity. Educational systems must prepare workers for changing technical requirements. Research institutions must evaluate emerging tools and distinguish promising technologies from those that provide limited benefits. Extension and technical-service networks help producers understand how technologies perform under actual production conditions.

Technology should therefore be viewed as an expanding set of opportunities rather than a guaranteed source of productivity growth. The long-term contribution of artificial intelligence, robotics, precision agriculture, and other emerging technologies will depend on whether the broader agricultural system has the infrastructure, expertise, institutions, and financial capacity needed to incorporate them effectively.

Global Economic and Geopolitical Change

American agriculture operates within a global economy. International trade provides markets for U.S. agricultural products while exposing producers and businesses to changing foreign demand, competition, exchange rates, trade policy, transportation conditions, and geopolitical developments. Agricultural supply chains also depend on internationally connected markets for inputs and equipment.

The scale of this international connection is substantial. The United States exported approximately \$171 billion in agricultural goods in 2025, while agricultural imports reached approximately \$212 billion. The country's five largest agricultural export markets—Mexico, Canada, the European Union, Japan, and South Korea—accounted for 56 percent of U.S. agricultural exports that year (USDA ERS, 2026).

These relationships create both opportunities and vulnerabilities. Access to international markets allows U.S. producers to specialize in commodities for which they are competitive and provides additional sources of demand. At the same time, concentrated dependence on particular markets can expose producers to disruptions when trade policy or geopolitical conditions change. The decline in U.S. agricultural exports to China in 2025, for example, illustrates how changes in trade relationships can alter commodity-specific market conditions (USDA ERS, 2026).

Agricultural imports are similarly important. Mexico, Canada, and the European Union together accounted for nearly 58 percent of U.S. agricultural import value during 2021–25, reflecting the degree to which the U.S. food system relies on international sources for certain commodities and processed foods (USDA ERS, 2026).

Input markets provide another example. USDA ERS reports that U.S. fertilizer production and consumption occur within a global fertilizer market and that the U.S. shares of global fertilizer production, imports, and exports have each declined substantially since 2006. Fertilizer markets are consequently influenced by international production, trade, and price conditions as well as domestic factors (Williams, Collins, & Boline, 2025).

Globalization also creates opportunities for scientific collaboration. Agricultural researchers increasingly exchange knowledge, genetic resources, technologies, and research methods across national boundaries. These relationships expand the pool of scientific knowledge available to researchers and allow agricultural systems in different countries to learn from one another.

The implications of geopolitical change therefore extend well beyond trade. Changes in diplomatic relationships, tariffs, transportation networks, international investment, supply chains,

or access to scientific and technological resources can influence agricultural capacity indirectly by changing the incentives and constraints facing producers and businesses.

These developments also reinforce the importance of flexibility. Because individual geopolitical events are difficult to anticipate decades in advance, agricultural resilience depends less on predicting specific events than on maintaining sufficient diversity and adaptability to respond when they occur. Diverse export markets, multiple suppliers, resilient transportation systems, strong domestic production, and continuing scientific collaboration can all reduce dependence on any single external condition.

The goal should not be economic isolation. International integration provides substantial benefits to American agriculture. The more durable objective is a system capable of participating in global markets while remaining sufficiently resilient to absorb disruptions within those markets.

Resource Competition and Land Use

Agriculture shares many of its most important resources with other sectors of the economy.

Land, water, energy, labor, and financial capital can all be directed toward competing uses. As population, technology, and economic conditions change, the relative value of those uses changes as well.

Land provides the clearest example. Agricultural land supports food and fiber production, but the same landscapes may also be used for housing, energy development, transportation, industry, recreation, and conservation. USDA ERS tracks these competing uses through its land-use and land-cover assessments because changes in land conversion can affect the availability and distribution of agricultural land over time.

The scale of land-use change is significant. USDA ERS estimates that between 2001 and 2016, approximately 16.0 million acres of agricultural land were converted to developed land uses, while millions of additional acres changed among agricultural, forest, and other uses.

Development pressure has been particularly concentrated near existing population centers and areas experiencing population growth.

The implications are not limited to the total number of acres converted. Location matters.

Conversion of highly productive agricultural land or fragmentation of agricultural areas can affect agricultural operations differently from conversion of marginal land in regions with abundant agricultural acreage. The continuity of land resources can matter for infrastructure, irrigation districts, local supply chains, and the viability of agricultural communities as well.

Water is another resource shared among competing users. Agricultural, municipal, industrial, recreational, and environmental demands can draw upon the same watersheds and aquifers. As discussed earlier, technological improvements can increase agricultural output per unit of water, but efficiency improvements do not eliminate the underlying competition for finite supplies.

Energy, labor, and capital likewise move among sectors in response to changing returns. Higher returns in one part of the economy can increase competition for workers and investment capital, while changes in energy prices can affect the cost of irrigation, transportation, fertilizer production, processing, and machinery.

Resource competition should not therefore be understood as an entirely zero-sum process.

Productivity improvements can allow more output to be produced from existing resources, while technological innovation can change the economic value of competing uses. Better irrigation technology, for example, can increase agricultural production without requiring a proportional

increase in water withdrawals. Similarly, advances in agricultural productivity can reduce pressure for land conversion by allowing existing farmland to produce more output.

At the same time, productivity improvements cannot eliminate resource constraints entirely. Land, water, energy, labor, and capital all have alternative uses, and decisions made today can affect agricultural options decades into the future. Long-term planning should therefore account for both the immediate economic value of resources and the value of preserving productive options for future generations.

Strategic Uncertainty and Adaptive Capacity

The future operating environment of agriculture cannot be predicted with precision. Scientific discoveries, technological breakthroughs, demographic change, environmental variability, market conditions, geopolitical developments, and unforeseen disruptions will continue to reshape agricultural systems in ways that cannot be anticipated in detail decades in advance.

This uncertainty is not simply a temporary limitation in current knowledge. It is an inherent feature of long-term planning. The technologies that will dominate agriculture in 2076 cannot be identified with confidence today, just as previous generations could not have anticipated the full consequences of many technologies now considered routine.

The inability to predict specific future conditions does not, however, prevent effective preparation. Agricultural systems are better positioned to withstand change when they possess strong underlying capabilities rather than narrowly optimized responses to a single scenario. Productive soils, reliable infrastructure, scientific research, education, financial resilience,

diverse markets, and effective institutions all increase the range of responses available when circumstances change.

Adaptive capacity is therefore itself a form of strategic capital. Rather than attempting to anticipate every future challenge individually, agricultural policy can strengthen capabilities that make adaptation easier: better information, stronger research systems, effective knowledge transfer, financial flexibility, modern infrastructure, diversified production, and healthy natural resources.

The historical record supports this approach. Major transformations such as mechanization, improved genetics, synthetic fertilizers, irrigation, precision agriculture, and digital technologies emerged under different circumstances. Their contribution to agricultural productivity depended not simply on the existence of the technologies, but on whether producers and institutions could evaluate them, invest in them, adapt them, and incorporate them into existing production systems. The history of technology adoption therefore reinforces the broader conclusion that agricultural resilience depends on the capacity to learn rather than on technological sophistication alone (McFadden et al., 2023; USDA ERS, 2026).

Long-term planning should consequently emphasize capabilities that remain valuable across multiple plausible futures. Investments in scientific research, education, infrastructure, natural resource stewardship, institutional effectiveness, and economic resilience are unlikely to lose their value simply because future conditions differ from current expectations.

This approach also reduces dependence on any single forecast. Rather than attempting to determine exactly what American agriculture will look like several decades from now,

policymakers can focus on preserving the conditions that allow producers and institutions to respond as circumstances unfold.

In an environment characterized by uncertainty, the ability to learn, adjust, and continue investing may ultimately be more important than the ability to predict precisely what comes next.

Synthesis of Long-Term Drivers

The long-term drivers examined in this chapter differ in their origins, scale, and pace, but they share an important characteristic: they change the conditions under which agriculture operates rather than eliminating the fundamental importance of the capabilities that support agricultural productivity.

Environmental change can alter production conditions and resource availability. Technological acceleration can expand the range of tools available to producers. Globalization can open new markets while increasing exposure to international disruptions. Demographic change can reshape labor availability, succession, and rural communities. Competition for resources can influence the amount and cost of land, water, labor, energy, and capital available to agriculture.

These forces also interact. Climate conditions can influence demand for irrigation technologies and crop genetics. Demographic change can influence labor costs and incentives for automation. Geopolitical developments can affect both agricultural exports and the availability of production inputs. Resource constraints can encourage technological innovation, while technological advances can change the economics of resource use.

The combined effect is therefore more important than any individual driver considered in isolation. Long-term agricultural planning must account for these interactions rather than attempting to manage each trend independently.

The strongest conclusion from this analysis is that resilience depends more on capability than on prediction. Agricultural systems cannot eliminate uncertainty, and policymakers cannot anticipate every technological, environmental, economic, or geopolitical development that will occur during the next several decades. They can, however, strengthen the ability of producers and institutions to respond effectively when conditions change.

This principle has become increasingly important as the operating environment grows more interconnected. Agricultural production depends on global markets, complex supply chains, sophisticated technologies, specialized workforces, and natural resources that may respond slowly to changing conditions. The more interconnected the system becomes, the more valuable flexibility and redundancy become.

Long-term policy should therefore favor investments that preserve options. Scientific research creates future technological choices. Education and workforce development create human capacity. Infrastructure provides physical flexibility. Natural resource stewardship preserves the biological foundation for future production. Financial resilience gives producers room to adapt. Diverse markets and production systems reduce dependence on individual pathways.

The purpose of long-term agricultural policy should ultimately be to ensure that these capabilities remain strong enough for future generations to respond to circumstances that cannot yet be predicted.

Chapter Findings

Finding 1: Long-term agricultural performance will be shaped by the interaction of environmental, technological, demographic, economic, and geopolitical forces rather than by any single external driver (USDA Climate Hubs, 2026; USDA ERS, 2026).

Finding 2: Environmental change is likely to increase the importance of adaptive management, resource stewardship, scientific research, and technologies that improve agricultural resilience across different regional conditions (USDA Climate Hubs, 2026).

Finding 3: Technological acceleration expands agricultural opportunities but does not guarantee productivity gains; adoption depends on infrastructure, technical expertise, financial capacity, and the demonstrated value of individual technologies (McFadden et al., 2023).

Finding 4: Global markets create substantial opportunities for American agriculture while also exposing producers and supply chains to international competition, trade-policy changes, and disruptions affecting agricultural inputs and markets (USDA ERS, 2025, 2026; Williams et al., 2025).

Finding 5: Competition for land, water, labor, energy, and capital can influence agricultural capacity over long planning horizons, making productivity improvements and resource stewardship important complements to efforts to preserve the agricultural resource base.

Finding 6: Adaptive capacity is a strategic capability that allows agriculture to respond to uncertain future conditions without relying on a single forecast or technological pathway.

The analytical portion of this assessment has now identified the principal foundations of long-term agricultural capacity and the major forces that may reshape the environment in which those foundations operate. The remaining task is to determine how policy can strengthen those capabilities while preserving the flexibility needed to respond to future change.

The following chapter brings the evidence together into a strategic framework for long-term agricultural capacity, establishing the principles that will guide the specific policy recommendations presented later in the assessment.

A Strategic Framework for Long-Term Agricultural Capacity

Introduction

The preceding chapters examined the major foundations of long-term agricultural capacity: innovation, natural capital, human capital, physical infrastructure, economic institutions, and demographic continuity. Although these areas involve different disciplines and policy responsibilities, the evidence points toward a common conclusion. Agricultural productivity is sustained not by any single resource or institution, but by the interaction among several forms of productive capital that must be maintained and renewed over time (USDA ERS, 2026; Wang et al., 2012, 2015).

This conclusion has important implications for policy. Productive soils are valuable, but their contribution depends partly on scientific knowledge and effective management. Research can generate new technologies, but those technologies require skilled workers, infrastructure, financing, and functioning markets before they can produce broad gains. Likewise, strong

markets cannot compensate indefinitely for degraded natural resources or declining scientific and institutional capacity.

The purpose of this chapter is therefore not to introduce another set of empirical findings. Instead, it brings the evidence together into a strategic framework for long-term policy. The principles that follow describe the characteristics of an agricultural system capable of remaining productive, economically viable, and adaptable over several decades. They are intentionally broader than individual programs or legislative proposals. Their purpose is to provide a basis for evaluating specific policy choices.

Principle 1

Preserve Agricultural Capacity Before Expanding It

The first priority should be protecting the productive assets agriculture already possesses while continually improving how those assets are used. Productive soils, reliable water systems, scientific institutions, infrastructure, and accumulated human expertise often represent investments built over decades or generations. Once these assets have been substantially degraded or lost, restoring them may require far more time and resources than maintaining them in the first place (USDA NRCS, 2026; USDA ARS, 2026; USDA ERS, 2026).

The historical record makes the importance of this principle clear. Agricultural productivity has increased substantially without requiring comparable growth in traditional production inputs, but those productivity gains have depended upon maintaining the natural, scientific, human, and physical foundations that make continued improvement possible (USDA ERS, 2026; Wang et al., 2015).

Preservation should not be confused with resisting change. American agriculture has advanced precisely because it has continually changed its technologies, production practices, business structures, and management systems. The objective is not to preserve farming exactly as it exists today. It is to preserve the underlying capacity for productive agriculture while allowing the methods used to achieve that production to evolve.

This principle also reflects an important distinction between maintenance and expansion. Policy discussions often focus on creating new productive opportunities, but preventing deterioration of existing productive assets can be equally consequential. Soil degradation, declining infrastructure, loss of scientific capacity, or erosion of institutional knowledge may occur gradually and therefore attract less attention than new investments. Yet preventing those losses can preserve opportunities that would otherwise become increasingly expensive to recover.

Long-term agricultural policy should consequently treat preservation as an active form of investment. Maintaining the productive base does not mean avoiding change; it means ensuring that change takes place without unnecessarily reducing the options available to future generations.

Principle 2

Invest Across Complementary Systems

Agricultural productivity emerges from interactions among several forms of capital rather than from isolated investments. Research generates new knowledge, but educated professionals are needed to apply it. Infrastructure expands what producers can accomplish, but skilled personnel and effective institutions are needed to operate it. Natural resources support production most

effectively when combined with appropriate technologies and sound management. Markets allocate resources more effectively when producers have access to reliable transportation, information, financing, and communications systems (Wang et al., 2012; USDA ERS, 2026).

This interconnectedness means that policy evaluation should extend beyond the performance of individual programs. An investment can generate greater value when complementary systems are functioning well. Broadband infrastructure, for example, is more valuable when producers have access to precision technologies and the technical expertise needed to use them. Research investments are more productive when educational institutions are preparing scientists and technicians and when Extension and other knowledge-transfer systems can connect new findings with producers (McFadden et al., 2023; NIFA, 2025).

The reverse is also true. Weakness in one component can constrain the value of investments elsewhere. Advanced technology has limited practical value when infrastructure is inadequate. Productive farmland is less valuable when water systems or market access are unreliable. Research capacity is less effective when the workforce needed to sustain it is shrinking.

This suggests that long-term policy should emphasize coordination across research, education, infrastructure, natural resource management, and economic institutions rather than treating each area as an independent objective.

The objective is not necessarily to increase spending in every area simultaneously. It is to identify the combinations of investments that reinforce one another and produce durable improvements in the performance of the agricultural system as a whole.

Principle 3

Strengthen Adaptive Capacity

Agricultural history demonstrates that change is inevitable even when its exact form cannot be predicted. Environmental conditions change, technologies evolve, demographic patterns shift, markets adjust, and geopolitical relationships develop in unexpected ways. The agricultural systems that have remained productive over time have generally been those capable of adapting to such changes rather than those that remained optimized for a single set of conditions (USDA Climate Hubs, 2026; USDA ERS, 2026).

Adaptive capacity is therefore itself a productive asset. Research expands the range of technological responses available to producers. Education and Extension allow new knowledge to be incorporated into practice. Infrastructure creates operational flexibility, while diversified production and financial resilience reduce dependence on individual pathways. Together, these capabilities give producers and institutions more options when circumstances change.

This principle is particularly important because long-term agricultural planning cannot reliably depend upon a single forecast. USDA research on technology adoption, for example, demonstrates that the usefulness and adoption of new technologies depend on their costs, supporting infrastructure, technical requirements, and local production conditions (McFadden et al., 2023). Likewise, the effects of environmental change differ across regions and production systems (USDA Climate Hubs, 2026).

A resilient system is therefore not one that is perfectly optimized for an expected future. It is one that can perform reasonably well across a range of plausible futures and adjust when new information becomes available.

Adaptive capacity should consequently be treated as a continuing investment rather than something reserved for emergencies. Institutions that support experimentation, learning, knowledge exchange, and technological adoption make it possible for agriculture to adjust incrementally instead of relying primarily on crisis-driven responses.

Principle 4

Support Long-Term Decision Making

Agriculture operates on time horizons that often extend beyond a single growing season, business cycle, or political term. Soil conservation, irrigation systems, orchards, breeding programs, research infrastructure, workforce development, and major capital improvements may require years or decades before their full benefits become apparent (USDA ERS, 2026; USDA ARS, 2026).

This creates an important challenge for policy. Short-term indicators such as annual production, farm income, or commodity prices are useful for understanding current conditions, but they do not necessarily reveal whether the underlying productive capacity of agriculture is strengthening or deteriorating.

A policy can produce favorable short-term results while gradually weakening the assets on which future production depends. The reverse is also possible: an investment may impose costs in the near term while generating benefits that accumulate over many years. Research and infrastructure provide obvious examples of this pattern.

Long-term decision making therefore requires evaluating not only immediate outcomes but also how current choices affect the productive options available to future generations. This does not mean ignoring short-term economic conditions or current producer needs. It means ensuring that responding to immediate pressures does not systematically reduce the ability to invest in future productivity.

This principle is particularly important for policies affecting natural resources and infrastructure because those assets often have long service lives. Soil degradation, groundwater depletion, deferred maintenance, and the loss of research capacity may occur incrementally, while their consequences can persist for decades.

An effective policy framework should therefore reward decisions that maintain productive assets while allowing them to evolve. The goal is to balance current needs with the preservation and improvement of the foundations required for future production.

Principle 5

Maintain Scientific and Institutional Capacity

Sustained agricultural productivity depends on institutions capable of generating, evaluating, preserving, and communicating knowledge. Universities, research agencies, Extension organizations, producer groups, private firms, and government institutions perform different functions within this system, but their combined capacity allows agriculture to continue learning and adapting (Fuglie & Heisey, 2007; NIFA, 2025; USDA ERS, 2026).

Scientific capacity is cumulative. Long-term field experiments, germplasm collections, research databases, monitoring networks, breeding programs, and specialized expertise often take years or decades to build. USDA's Long-Term Agroecosystem Research Network illustrates the value of maintaining research programs across decades, while USDA germplasm collections preserve biological resources that may become important for future breeding and scientific work (USDA ARS, 2026).

Institutional continuity matters for similar reasons. Organizations retain knowledge about research programs, management practices, regulatory systems, and past responses to emerging problems. When that knowledge is lost, future institutions may need to reconstruct information that previously took years to develop.

Collaboration is equally important. Agricultural problems rarely conform neatly to disciplinary or organizational boundaries. Effective solutions often require researchers, producers, engineers, economists, educators, private firms, and public agencies to work together. Extension provides one established mechanism for connecting scientific knowledge with producers and communities, but similar relationships exist throughout the broader agricultural innovation system (NIFA, 2025).

Maintaining scientific and institutional capacity should therefore be understood as an investment in future options. Strong institutions do more than address the problems already known. They preserve the ability to investigate new questions, respond to unexpected challenges, and develop technologies and policies for circumstances that cannot yet be anticipated.

Principle 6

Recognize Agriculture as a National Strategic Asset

Agriculture performs functions extending well beyond commodity production. It contributes to food security, rural economies, scientific research, environmental stewardship, industrial supply chains, international trade, and national economic activity. The capacity to maintain productive agriculture therefore contributes to several dimensions of national resilience simultaneously.

Recognizing agriculture as a strategic asset does not mean that agricultural production should be centrally directed or insulated from markets. Rather, it recognizes that certain agricultural capabilities generate benefits extending beyond individual transactions. Productive farmland, scientific institutions, transportation infrastructure, educational systems, and resilient food and agricultural supply chains can contribute to broader national interests even when those benefits are not fully reflected in individual market decisions.

The strategic importance of agricultural capacity is also illustrated by the sector's international role. The United States remains a major agricultural exporter, while also relying on international markets for important agricultural products and inputs (USDA ERS, 2026). Maintaining productive domestic capacity therefore matters not only to producers but also to the country's broader economic and supply-chain resilience.

This perspective also encourages coordination across policy areas. Transportation, energy, education, research, water management, trade, land use, and rural development decisions can all affect agriculture even when agriculture is not their primary objective. Treating agriculture as a cross-cutting national capability creates a stronger rationale for considering those interactions when making long-term investments.

Strategic assets are generally developed over long periods but can be weakened gradually through underinvestment or deferred maintenance. Preserving agricultural capacity therefore requires continued stewardship and periodic modernization rather than assuming that past investments will remain sufficient indefinitely.

Principle 7

Build Resilience Through Diversity

Resilient agricultural systems are rarely dependent upon a single crop, technology, market, transportation route, institution, or production model. Diversity across these areas provides alternatives when one component becomes less effective or is disrupted.

This principle appears repeatedly throughout the evidence examined in this assessment. Diverse production systems can reduce dependence on a single biological or environmental condition. Multiple transportation modes can provide alternatives when individual corridors are disrupted. Diverse markets can reduce exposure to the loss of a single destination. Multiple research institutions and technological pathways create additional opportunities for innovation.

Diversity does not necessarily mean abandoning specialization or efficiency. Highly specialized systems can be extremely productive under stable conditions. The concern is that a system optimized too narrowly for current conditions may become more vulnerable when those conditions change.

The value of diversity lies in preserving options. Maintaining multiple pathways for agricultural production, innovation, market access, and institutional support increases the likelihood that some components will remain effective when circumstances depart from historical norms.

This principle should consequently be distinguished from an argument for inefficiency.

Resilience requires some capacity that may appear redundant under normal conditions but becomes valuable during disruption. The challenge for policy is to preserve sufficient diversity and redundancy in strategically important systems without imposing unnecessary costs on producers or businesses.

Long-term agricultural resilience should therefore be measured not only by efficiency under normal conditions, but also by the range of effective responses available when conditions change.

Integrating the Strategic Principles

The seven principles presented in this chapter are closely connected rather than independent policy objectives. Preserving productive capacity creates a stronger foundation for long-term investment. Investments across complementary systems increase the effectiveness of one another. Scientific institutions generate the knowledge needed for continued innovation, while infrastructure and human capital allow those innovations to be implemented. Recognizing agriculture as a strategic national capability encourages coordination across policy areas, while diversity preserves flexibility when individual systems come under stress.

Together, these principles point toward an agricultural strategy built around capability rather than prediction. The objective is not to anticipate every future development with precision, but to

maintain the resources, institutions, knowledge, and flexibility necessary to respond when circumstances change.

This approach is consistent with the historical development of American agriculture.

Productivity growth has emerged through the accumulation of improvements in science, technology, management, infrastructure, education, and resource stewardship rather than from dependence on any single technological breakthrough (USDA ERS, 2026; Wang et al., 2015).

It also provides a useful standard for evaluating policy recommendations. Individual policies should not be judged solely by their immediate effects or by whether they address one identifiable problem. They should also be evaluated according to whether they strengthen the underlying capabilities that support agricultural productivity over time.

The most durable policies are therefore likely to be those that reinforce several capabilities simultaneously. An investment that strengthens research, for example, may produce greater long-term benefits when paired with workforce development, infrastructure, and knowledge-transfer institutions. Likewise, a natural resource investment may become more valuable when supported by research and technological innovation.

The strategic framework consequently provides a bridge between the analytical findings of this assessment and the specific policy recommendations that follow. The recommendations should be understood not as isolated initiatives, but as practical applications of the broader principles developed throughout the report.

Chapter Findings

Finding 1: Long-term agricultural capacity is strengthened when foundational productive assets are preserved while their performance continues to improve (USDA NRCS, 2026; USDA ERS, 2026).

Finding 2: Coordinated investment across research, education, infrastructure, natural resources, markets, and institutions can produce greater long-term benefits than isolated interventions (Wang et al., 2012; McFadden et al., 2023).

Finding 3: Adaptive capacity is itself a strategic asset because it allows agricultural systems to respond when future conditions differ from expectations (USDA Climate Hubs, 2026).

Finding 4: Scientific institutions, organizational continuity, and sustained investment are essential to maintaining agricultural productivity across generations (Fuglie & Heisey, 2007; NIFA, 2025; USDA ARS, 2026).

Finding 5: Agriculture represents a strategic national capability whose productive foundations warrant sustained stewardship and coordination across policy domains (USDA ERS, 2026).

Finding 6: Diversity across production systems, technologies, markets, infrastructure, and institutions strengthens resilience by expanding the range of effective responses available when conditions change.

The framework developed here provides the conceptual bridge between the analytical findings and the policy recommendations that follow. The next chapter translates these principles into specific recommendations for strengthening United States agricultural capacity over the next twenty-five to fifty years.

Strategic Policy Recommendations

Introduction

The preceding chapters identified the major foundations of long-term agricultural capacity and brought those findings together into a strategic framework for policy. This chapter translates that framework into recommendations intended to preserve and strengthen the productive foundations of United States agriculture over the next twenty-five to fifty years.

The recommendations are organized around capabilities rather than existing agencies or congressional committee structures. That distinction is deliberate. Agricultural capacity emerges from interactions among natural resources, scientific research, infrastructure, markets, education, workforce development, and governance. No single federal agency controls all of these systems, and policies developed in one area can affect the performance of others (Wang et al., 2012; USDA ERS, 2026).

The recommendations should therefore not be understood as a fixed legislative agenda. Different administrations and Congresses may pursue the same objectives through different statutes, appropriations, regulations, or administrative programs. The purpose here is to identify durable priorities that remain relevant as political, economic, environmental, and technological conditions change.

Recommendation 1

Establish a National Agricultural Capacity Strategy

Recommendation

The United States should establish a long-term national strategy focused explicitly on preserving and strengthening agricultural capacity over a twenty-five- to fifty-year planning horizon.

Rationale

Agricultural policy necessarily responds to immediate conditions. Commodity prices fluctuate, severe weather can disrupt production, disease outbreaks can emerge unexpectedly, and annual appropriations influence the resources available to federal agencies. Those short-term responsibilities are unavoidable, but they do not replace the need for long-term planning.

The productive foundations of agriculture develop over much longer periods. Research institutions, scientific expertise, productive soils, water infrastructure, transportation networks, and agricultural businesses all depend upon sustained investment and institutional continuity (USDA ERS, 2026; USDA ARS, 2026). The productivity record itself illustrates why this matters: U.S. agricultural productivity has increased over many decades through cumulative improvements in research, technology, management, and resource use rather than through a single intervention (USDA ERS, 2026).

A national agricultural capacity strategy would provide a framework for maintaining continuity across changes in political leadership while periodically assessing the condition of the nation's agricultural capabilities. Its purpose should not be to predict exactly what American agriculture will look like decades from now. Instead, it should identify the capabilities that need to remain strong across a range of plausible futures.

The strategy should monitor trends affecting scientific research, natural resources, workforce development, infrastructure, technological innovation, food systems, market access, and

international competitiveness. It should also identify areas in which declining capacity could create significant long-term risks.

Coordination across federal agencies should be an important part of the strategy. Transportation, energy, education, water resources, scientific research, rural development, and trade decisions can all affect agricultural performance even when agriculture is not their primary focus. The interdependence documented throughout this assessment suggests that evaluating these systems separately can obscure important connections among them.

The strategy should therefore function as a mechanism for long-term assessment and coordination, not as a detailed central plan for agricultural production. Periodic reassessment would allow priorities to change as conditions change while maintaining a consistent focus on preserving national agricultural capacity.

Expected Long-Term Benefits

- Greater continuity in agricultural policy and investment across administrations.
- Better coordination among federal institutions whose decisions affect agriculture.
- More consistent identification of long-term investment needs.
- Earlier recognition of emerging weaknesses and opportunities.
- Greater preparedness for environmental, technological, economic, and geopolitical change.

Recommendation 2

Sustain Public Investment in Agricultural Research and Innovation

Recommendation

Federal and state governments should maintain sustained investment in agricultural research, scientific infrastructure, and innovation systems while strengthening collaboration among universities, producers, nonprofit organizations, and the private sector.

Rationale

The historical productivity record makes the importance of innovation difficult to ignore. USDA ERS estimates that total factor productivity accounted for most U.S. agricultural output growth between 1948 and 2023, with TFP increasing at an average annual rate of 1.40 percent over that period (USDA ERS, 2026). Research and development have been identified as important contributors to the technical change underlying long-term productivity growth (Fuglie & Heisey, 2007; Wang et al., 2013).

Scientific capacity, however, cannot be created immediately when a new challenge appears. Research facilities, long-term field experiments, breeding programs, germplasm collections, monitoring networks, and specialized scientific expertise often require years or decades to develop. The USDA Agricultural Research Service's Long-Term Agroecosystem Research Network illustrates this cumulative character, with research sites possessing decades of observations that support long-term analysis of agricultural systems (USDA ARS, 2026).

Public investment is particularly important for areas of research whose benefits are broad, uncertain, or difficult for individual firms to capture. Economic research has found substantial returns to public agricultural research, while USDA analyses indicate that public and private

R&D perform complementary roles within the broader innovation system (Fuglie & Heisey, 2007; USDA ERS, 2019).

Public research should therefore not be viewed as a substitute for private innovation. The two sectors perform different but reinforcing functions. Universities and public laboratories can sustain foundational research and long-term data collection, while private firms can develop, commercialize, and refine technologies in response to market demand.

The objective should be to maintain an innovation ecosystem capable of generating solutions that cannot yet be anticipated. Sustained research investment is consequently an investment in future options rather than simply an effort to produce immediate technological breakthroughs.

Expected Long-Term Benefits

- Continued productivity and resource-use improvements.
- Greater capacity to respond to emerging agricultural challenges.
- Preservation of long-term scientific datasets and research infrastructure.
- Faster evaluation and refinement of promising technologies.
- Stronger connections among public research, producers, and private innovators.

Recommendation 3

Protect and Enhance Productive Natural Capital

Recommendation

Agricultural policy should prioritize the long-term stewardship of soils, water resources, biodiversity, and other renewable natural resources that form the foundation of agricultural production.

Rationale

Natural capital is productive capital. Soils provide physical support, water storage, nutrients, and biological functions; water systems determine where and how reliably agricultural production can occur; and biodiversity supports processes such as pollination, pest regulation, nutrient cycling, and genetic improvement (USDA NIFA, 2025; USDA NRCS, 2026).

The long-term nature of these resources makes prevention particularly important. Soil degradation, groundwater depletion, habitat loss, and declining ecosystem function can develop gradually while their consequences persist for decades. USDA NRCS emphasizes soil health and conservation practices as means of maintaining soil function and agricultural productivity, while USDA and USGS data demonstrate the importance of sustainable water management to irrigation-dependent agriculture (USDA NRCS, 2026; USGS, 2024).

The economic case for stewardship is therefore inseparable from the environmental case. Healthy soils can support continued production while improving water infiltration and nutrient cycling. Efficient water management can increase output per unit of water while reducing pressure on limited supplies. Biodiversity and ecosystem functions can provide services that support production and preserve options for future adaptation (USDA Climate Hubs, 2026).

Policy should consequently emphasize both protection and improvement. Agricultural systems should be encouraged to increase productivity while maintaining the natural resource base on

which that productivity depends. This approach does not require choosing between production and stewardship. Properly designed, stewardship can strengthen the long-term economic viability and resilience of agricultural systems.

Expected Long-Term Benefits

- Sustained or improved agricultural productivity.
- Greater resilience to drought and environmental variability.
- More efficient use of land and water resources.
- Reduced long-term costs associated with resource degradation.
- Preservation of productive opportunities for future generations.
- Stronger ecological foundations for continued agricultural innovation.

Recommendation 4

Modernize Agricultural Infrastructure

Recommendation

Public and private investment should prioritize the maintenance, modernization, and strategic expansion of transportation, water management, energy, communications, storage, processing, and research infrastructure that supports agricultural production.

Rationale

Infrastructure is productive capital. Transportation networks connect producers with suppliers and markets; water systems improve production reliability; energy and communications support increasingly sophisticated technologies; storage and processing facilities preserve and increase

the value of agricultural products; and research infrastructure provides the physical foundation for continued scientific advancement (USDA AMS, 2020, 2026; USDA ERS, 2026; USDA ARS, 2026).

The interconnected nature of this infrastructure means that individual investments can generate benefits beyond the assets themselves. USDA agricultural transportation analyses demonstrate the importance of major highway, rail, waterway, and port networks to agricultural freight movement, while research on precision agriculture identifies broadband and related infrastructure as important enabling conditions for digital technology adoption (USDA AMS, 2020; McFadden et al., 2023).

Maintenance is particularly important. Roads, bridges, irrigation systems, storage facilities, electrical networks, and research facilities all deteriorate over time and require continuing investment. Deferred maintenance can eventually undermine the productivity gains that infrastructure was intended to provide.

Infrastructure planning must also anticipate technological change. Precision agriculture, automation, remote sensing, artificial intelligence, and increasingly data-intensive production systems all depend on reliable supporting infrastructure. The useful life of an infrastructure investment therefore depends partly on its ability to accommodate technologies that may not yet be widely adopted.

Modernization should consequently be treated as an ongoing process rather than a one-time construction effort. Maintaining existing systems while upgrading them to meet emerging requirements can strengthen both normal operating efficiency and resilience during disruptions.

Expected Long-Term Benefits

- Lower production, transportation, and logistics costs.
- Greater resilience throughout agricultural supply chains.
- More reliable water, energy, and communications systems.
- Faster adoption of emerging agricultural technologies.
- Greater operational flexibility across regions.
- Reduced risks associated with aging infrastructure.

Recommendation 5

Strengthen Agricultural Education and Workforce Development

Recommendation

Federal, state, and educational institutions should strengthen agricultural education, workforce development, and professional training across the full range of agricultural occupations while expanding opportunities for continued learning throughout agricultural careers.

Rationale

The long-term capacity of American agriculture depends heavily on the people capable of operating, maintaining, and improving increasingly sophisticated production systems. USDA ERS research shows that agricultural labor has declined substantially in quantity while labor quality has increased, with changes in education and other worker characteristics helping offset some of the effects of declining labor inputs on agricultural production (Wang et al., 2022; Wang & Njuki, 2025).

Modern agriculture increasingly requires expertise in biology, engineering, veterinary medicine, environmental science, economics, data analysis, automation, robotics, and information technology. USDA NIFA's workforce-development programs likewise identify the preparation of scientists, educators, Extension professionals, and other agricultural specialists as an important component of national agricultural capacity (NIFA, 2026).

Education should therefore be treated as productive infrastructure. Universities, community colleges, technical programs, apprenticeships, internships, Extension services, and continuing education provide different pathways into the agricultural workforce and help professionals maintain relevant skills as production technologies change.

Formal education is only part of the process. Experienced farmers, ranchers, technicians, and managers possess practical knowledge developed through years of working under real production conditions. Mentorships, apprenticeships, producer networks, and other forms of experiential learning can preserve that knowledge while allowing newer participants to introduce updated scientific and technical capabilities.

Workforce development should consequently serve two purposes at once: preparing agriculture for technological change and preserving the practical expertise accumulated through previous generations. Strong relationships among educational institutions, producers, employers, and professional organizations can help ensure that training reflects actual agricultural needs.

Expected Long-Term Benefits

- A larger and more technically capable agricultural workforce.
- Faster and more effective adoption of emerging technologies.

- Stronger succession and knowledge-transfer pathways.
- Greater institutional continuity.
- Improved ability to adapt to scientific, technological, and environmental change.

Recommendation 6

Encourage Resilient and Diversified Agricultural Systems

Recommendation

Agricultural policy should encourage production systems that strengthen long-term resilience through diversification, flexibility, and effective risk management while recognizing the substantial differences among agricultural regions and production systems across the United States.

Rationale

No single production model is appropriate for every agricultural region. Climate, soils, water availability, markets, infrastructure, labor, and production objectives vary substantially across the United States. Policies intended to strengthen resilience should therefore allow producers to adapt strategies to local circumstances rather than prescribing a uniform model.

The evidence reviewed throughout this assessment suggests that diversity can provide resilience by reducing dependence on individual resources, technologies, markets, and transportation pathways. The same principle applies to individual agricultural businesses: producers can use combinations of crop and livestock diversification, marketing arrangements, insurance,

technology, water management, and other strategies to manage the risks they actually face (USDA ERS, 2025; Baldwin et al., 2026).

This does not mean that diversification is always economically preferable to specialization.

Highly specialized systems can be extremely efficient under appropriate conditions. The policy objective should instead be to preserve sufficient flexibility for producers to adapt when those conditions change. Public policy can support that flexibility through access to research, information, technical assistance, financial tools, infrastructure, and markets. Producers should retain substantial discretion over which combination of practices best fits their circumstances.

Regional diversity itself is also valuable. Different agricultural regions possess different natural resources, expertise, infrastructure, and production capabilities. Maintaining that diversity reduces dependence upon a limited number of production systems while preserving the advantages that arise from regional specialization.

Resilience should therefore be measured not only by performance under normal conditions, but also by the ability to adjust when environmental, economic, technological, or market conditions change.

Expected Long-Term Benefits

- Greater resilience to environmental and economic change.
- Reduced dependence on individual crops, technologies, resources, or markets.
- Greater flexibility for producers.
- Stronger capacity to absorb and recover from disruptions.
- Preservation of regional agricultural diversity.

- Greater opportunity for continued innovation and adaptation.

Recommendation 7

Improve Strategic Coordination Across Agricultural Policy

Recommendation

Federal agricultural policy should strengthen coordination among agencies and institutions whose decisions influence long-term agricultural capacity, including those responsible for transportation, energy, water resources, education, research, environmental management, trade, and rural development.

Rationale

Agricultural productivity depends on a broader set of systems than those administered by traditional agricultural agencies alone. Transportation affects market access; energy influences irrigation, processing, fertilizer production, and equipment; water policy affects production reliability; education shapes workforce capability; research supports technological development; and trade policy influences export opportunities and supply chains.

The evidence throughout this assessment demonstrates that these systems are complementary. USDA's agricultural transportation research, for example, treats highways, railroads, waterways, and ports as interconnected components of agricultural supply chains (USDA AMS, 2020, 2026). Likewise, USDA research on technology adoption shows that infrastructure, education, technical support, and economic incentives can influence whether agricultural technologies are successfully adopted (McFadden et al., 2023).

A weakness in one system can therefore reduce the effectiveness of investment in another. Fragmented decision-making may overlook these relationships or create unintended conflicts among otherwise reasonable policy objectives.

Improved coordination does not require placing agricultural policy under a single institution. Agencies can retain their specialized responsibilities while establishing mechanisms for assessing the agricultural implications of major investments and policies and for coordinating where objectives overlap.

Coordination is especially important for long-lived investments. Transportation corridors, water systems, research institutions, educational programs, and energy infrastructure can influence agricultural opportunities for decades. Incorporating agricultural considerations into broader planning can help ensure that public investments reinforce rather than inadvertently constrain agricultural capacity.

Agriculture should consequently be treated as a cross-cutting national capability supported by multiple areas of public policy.

Expected Long-Term Benefits

- More effective use of public resources.
- Greater coherence among policies affecting agriculture.
- Reduced duplication and unintended conflicts.
- Stronger coordination among complementary institutions.
- Better alignment between agricultural priorities and broader national investment.

Recommendation 8

Institutionalize Long-Term Assessment

Recommendation

The United States should establish a recurring, independent assessment of national agricultural capacity to evaluate long-term trends, identify emerging risks and opportunities, and provide evidence-based guidance for policymakers.

Rationale

Agricultural capacity changes gradually. Soil condition, infrastructure, scientific capabilities, workforce demographics, technological adoption, environmental conditions, and market structures can all change substantially over periods much longer than an election cycle.

A recurring national assessment would provide a structured way to monitor these trends and distinguish short-term fluctuations from structural changes. This would complement existing USDA data and analytical programs by bringing evidence from agricultural science, economics, engineering, ecology, demography, environmental science, and related fields into a single assessment of national agricultural capacity.

Regular assessment could also strengthen institutional memory. Changes in political leadership and administrative priorities can make it difficult to maintain a consistent understanding of long-term trends. A recurring assessment would provide successive administrations and Congresses with a common analytical baseline for evaluating agricultural policy and investment.

The purpose should not be to predict precisely what agriculture will look like decades from now. The strategic framework developed in this assessment argues against that approach. Instead,

assessment should improve situational awareness by tracking the condition of foundational capabilities, identifying emerging risks and opportunities, and evaluating whether existing policies and institutions are strengthening or weakening long-term capacity.

The process should also allow priorities to change. A recurring assessment should not lock policymakers into a fixed long-term plan. It should provide the information necessary to revise investments as circumstances evolve while maintaining a consistent focus on agricultural capacity.

Expected Long-Term Benefits

- Better-informed policymaking.
- Earlier identification of emerging risks and opportunities.
- Greater continuity in long-term agricultural planning.
- Stronger institutional memory and situational awareness.
- Greater ability to adjust policy as conditions change.

Integrating the Recommendations

Although each recommendation addresses a distinct aspect of agricultural policy, they should not be treated as independent initiatives. The analysis throughout this assessment demonstrates that agricultural capacity emerges from interactions among complementary forms of capital rather than from any single program or institution.

Consider agricultural research. Research can generate important scientific advances, but those advances produce their greatest value when educational institutions prepare skilled professionals, infrastructure allows technologies to be deployed, financial systems support investment, and

natural resources remain productive enough to sustain their use. The same relationship works in reverse: better infrastructure becomes more valuable when producers have the expertise to use it effectively, and scientific research can further increase the value of infrastructure by developing new technologies and management practices.

The recommendations therefore focus on building and preserving capability rather than prescribing a single model for American agriculture. Their purpose is to strengthen the foundations that allow agricultural systems to remain productive while adapting to changing conditions.

That distinction is important because the specific challenges agriculture will face over the next several decades cannot be known with precision. Environmental conditions, technologies, markets, demographics, and geopolitical circumstances will continue to evolve.

The most durable policy approach is consequently one that expands the range of responses available to future producers and institutions. Strong research institutions, productive natural resources, modern infrastructure, skilled professionals, functioning markets, diverse production systems, and effective coordination all increase that range of options.

This emphasis on capability reflects one of the central conclusions of the entire assessment: Resilience depends less on predicting the future than on maintaining the capacity to respond to it.

Chapter Findings

Finding 1: Long-term agricultural policy should prioritize preservation and renewal of the productive assets on which future agricultural performance depends.

Finding 2: Sustained public research, education, infrastructure, natural-resource stewardship, and financial capacity reinforce one another and should be considered as components of an integrated agricultural system.

Finding 3: Resilience is strengthened when producers and institutions retain flexibility in production, investment, technology adoption, and risk management.

Finding 4: Strategic coordination and recurring assessment can help policymakers identify interactions among agricultural systems and adjust priorities as conditions change.

The recommendations presented in this chapter translate the analytical findings of the assessment into a practical framework for long-term action. Taken together, they are intended not to predict the agricultural future, but to ensure that future generations inherit a system with sufficient resources, knowledge, infrastructure, institutional capacity, and flexibility to meet it.

Conclusions

This assessment began with a straightforward but consequential question:

What must the United States do to preserve and strengthen its agricultural capacity over the next twenty-five to fifty years?

The question is broad, encompassing agricultural science, economics, ecology, engineering, infrastructure, demographics, markets, and public policy. Yet the evidence reviewed throughout the assessment points toward a consistent conclusion. The future of American agriculture will depend less upon any single technological breakthrough, legislative initiative, or market

development than upon the nation's ability to maintain and strengthen the interconnected systems that support agricultural production (USDA ERS, 2026; Wang et al., 2012, 2015).

Agricultural capacity is not a fixed resource that can simply be preserved once and then relied upon indefinitely. It is a dynamic capability that has to be maintained, renewed, and improved. Productive soils, reliable water resources, scientific knowledge, skilled professionals, physical infrastructure, functioning markets, and effective institutions all contribute to that capability. None is sufficient on its own. Their value comes largely from the way they reinforce one another.

The historical record makes this cumulative character particularly clear. U.S. agricultural output has increased dramatically since 1948 while aggregate input use has grown much more slowly, with total factor productivity accounting for most of the growth in output (USDA ERS, 2026). That improvement has depended upon generations of investment in research, technology, education, infrastructure, resource management, and agricultural institutions (Fuglie & Heisey, 2007; Wang et al., 2015).

The same principle will shape the future. Maintaining agricultural capacity will require continued investment in the foundations that have supported productivity while allowing those foundations themselves to evolve. Protecting productive soils does not mean preserving farming practices unchanged. Maintaining scientific institutions does not mean restricting research to established technologies. Modernizing infrastructure does not mean abandoning existing systems. Supporting rural communities does not mean preventing demographic or economic change. In each case, the objective is to preserve the underlying capacity for continued improvement.

The assessment also demonstrates that resilience cannot be separated from adaptability. American agriculture has repeatedly adjusted to changes in technology, markets, labor, environmental conditions, and international economic relationships. Its ability to remain productive through those transitions has depended upon institutions and individuals capable of learning, experimenting, investing, and changing course when necessary (USDA Climate Hubs, 2026; USDA ERS, 2026).

This has an important implication for long-term policy. Because the specific conditions facing agriculture several decades from now cannot be known with confidence, policy should avoid excessive dependence upon any single forecast or technological pathway. Instead, long-term investments should strengthen capabilities that remain valuable across many plausible futures: scientific research, education, infrastructure, natural resource stewardship, financial resilience, institutional coordination, and reliable information.

The objective is therefore not simply to produce more. It is to preserve the capacity to produce, adapt, innovate, and recover as conditions change.

Principal Conclusions

Agricultural Capacity Is Cumulative

Perhaps the most important conclusion of this assessment is that agricultural capacity is built over time. Productive soils reflect decades of stewardship. Scientific knowledge grows through sustained research and experimentation. Infrastructure develops through long-term investment and maintenance. Educational institutions prepare successive generations of professionals, while agricultural experience and institutional knowledge are transmitted through producers,

universities, Extension organizations, professional networks, and other institutions (USDA ARS, 2026; NIFA, 2025).

These forms of productive capital do not operate independently. Better scientific knowledge can increase the effectiveness of infrastructure and resource management, while skilled workers are necessary to turn new technologies into practical improvements. Conversely, deterioration in one foundational area can gradually reduce the effectiveness of investments elsewhere.

This cumulative character changes the way agricultural policy should be evaluated. Capacity cannot be created or preserved through a handful of isolated initiatives. It requires continuing attention to the systems that support production and innovation. The benefits of that investment may not be visible immediately, but they compound over time.

The same is true of neglect. Declining infrastructure, degraded natural resources, loss of institutional expertise, or insufficient workforce development may develop gradually and therefore attract less attention than more immediate problems. Yet their eventual effects can be substantial and difficult to reverse (USDA NRCS, 2026; USDA ARS, 2026).

Preserving agricultural capacity should therefore be understood as an ongoing intergenerational responsibility. Each generation inherits productive assets created by those before it and, in turn, determines whether those assets will be strengthened, maintained, or allowed to deteriorate.

Innovation Remains the Primary Driver of Productivity

The history of American agriculture demonstrates the central importance of scientific and technological innovation to long-term productivity growth. USDA ERS estimates that total factor productivity increased at an average annual rate of 1.40 percent between 1948 and 2023, while

agricultural output grew much faster than aggregate input use over the same period (USDA ERS, 2026). Research has consistently identified technological change and agricultural research and development as major contributors to that long-term productivity growth (Fuglie & Heisey, 2007; Wang et al., 2013, 2015).

Advances in genetics, mechanization, agronomy, engineering, chemistry, information technology, automation, and biological science have repeatedly changed what producers can accomplish. Many of these advances have enabled more output from existing resources rather than simply expanding the quantity of those resources.

There is little reason to expect innovation to become less important in the future. Artificial intelligence, robotics, precision agriculture, advanced biotechnology, remote sensing, and other emerging technologies are already expanding the range of tools available to producers, although their eventual contribution will depend on adoption costs, infrastructure, technical expertise, and local production conditions (McFadden et al., 2023).

Technology alone, however, does not guarantee progress. New tools require research institutions to develop and evaluate them, workers capable of using them, infrastructure capable of supporting them, and producers able to incorporate them into practical decision-making. Innovation is therefore best understood as a **national capability**, not simply as a collection of individual technologies.

The central policy question is not which technology will dominate agriculture fifty years from now. It is whether the United States will retain the scientific, institutional, and human capacity necessary to develop and adopt whatever technologies prove most valuable.

Resilience Depends Upon Adaptation

Agricultural history makes one point particularly clear: change is unavoidable. Environmental conditions shift, technologies develop, consumer preferences evolve, international markets fluctuate, and geopolitical circumstances change. The agricultural systems that have remained successful have not been those that avoided change, but those that were able to adjust without losing their underlying productive capacity (USDA Climate Hubs, 2026; USDA ERS, 2026).

Adaptation depends upon more than the willingness of individual producers to change. It is supported by scientific research, education, diversified production systems, reliable infrastructure, effective institutions, financial resilience, and access to useful information. Together, these capabilities expand the range of responses available when conditions change.

This suggests that long-term agricultural policy should focus less on predicting exactly what the future will look like and more on ensuring that agriculture is capable of responding when that future arrives. No forecast can reliably anticipate every technological breakthrough, environmental disruption, market shift, or geopolitical development that may occur over several decades.

Resilience, in this sense, is not simply the ability to withstand disruption. It is the ability to absorb change, learn from it, and continue improving afterward. An agricultural system capable of adapting while preserving its productive foundations is better positioned for long-term success than one optimized narrowly for conditions that may not persist.

Agriculture Is a Strategic National Capability

Agriculture performs functions extending well beyond the production of food and fiber. It supports rural economies, food security, scientific research, environmental stewardship, industrial supply chains, and international trade. The scale of U.S. agricultural trade alone illustrates the sector's broader economic significance: the United States exported approximately \$171 billion in agricultural products in 2025 while importing roughly \$212 billion (USDA ERS, 2026).

Viewing agriculture as a strategic national capability therefore encourages a broader understanding of the policies that shape its future. Transportation, education, scientific research, energy, water management, land use, trade, and rural development decisions can all influence agricultural performance even when agriculture is not their primary objective.

Recognizing agriculture as strategic does not mean centrally directing production or insulating the sector from market forces. Rather, it recognizes that some of the foundations of agricultural capacity generate benefits that extend beyond individual farms and businesses. Productive land, scientific institutions, skilled professionals, resilient infrastructure, and reliable agricultural supply chains contribute to the broader national interest even where their benefits are not fully reflected in individual transactions.

Sustaining agricultural capacity will therefore require attention not only to what happens on farms and ranches, but also to the scientific, economic, environmental, infrastructural, and institutional systems surrounding them.

Looking Fifty Years Ahead

Trying to predict exactly what American agriculture will look like fifty years from now is neither realistic nor necessary. Previous generations could not have anticipated the full consequences of hybrid crops, synthetic fertilizers, satellite navigation, precision agriculture, genomic technologies, artificial intelligence, or autonomous machinery. There is little reason to believe the next generation will be able to anticipate all of the technologies that will matter in 2076.

What history does provide is a more useful lesson. Agricultural systems with strong scientific institutions, productive natural resources, skilled professionals, modern infrastructure, functioning markets, and adaptable governance have been better positioned to respond when conditions change (USDA ERS, 2026; USDA Climate Hubs, 2026).

The most effective long-term strategy is therefore not to prepare for one predicted future, but to strengthen the nation's ability to succeed across many plausible futures. That means preserving productive assets while improving their performance, maintaining strong scientific institutions, investing in education and workforce development, modernizing infrastructure, protecting natural resources, strengthening financial resilience, and supporting institutions capable of continual learning and adaptation.

In practical terms, this is a strategy of keeping options open. A country with strong research institutions can respond to problems it cannot yet foresee. A technically capable workforce can learn to use technologies that do not yet exist. Productive soils and reliable water systems give future producers a foundation upon which new production methods can be built. Modern infrastructure and functioning markets allow successful innovations to move beyond individual operations and into the broader agricultural economy.

The value of these investments therefore does not depend upon any particular forecast. Their purpose is to strengthen the underlying capacity of American agriculture to respond when circumstances change.

Final Assessment

The central finding of this report is straightforward:

The future strength of United States agriculture will depend less on the severity of any individual challenge than on the nation's ability to preserve and strengthen the productive foundations that allow agriculture to adapt, innovate, and remain economically viable.

Those foundations are well established. They include productive natural capital, strong scientific institutions, an educated and technically capable workforce, reliable physical infrastructure, functioning markets, resilient rural communities, and public institutions capable of supporting long-term investment while responding effectively to changing conditions.

None of these foundations can be taken for granted. Natural resources can be degraded, infrastructure can deteriorate, scientific capacity can weaken, institutional knowledge can be lost, and workforce shortages can emerge gradually. The danger is precisely that these problems may develop slowly enough to attract insufficient attention until the cost of correcting them becomes substantial (USDA NRCS, 2026; USDA ARS, 2026).

Preserving agricultural capacity is therefore not simply a matter of maintaining current production levels. It is about ensuring that future generations inherit the resources, knowledge, institutions, and opportunities necessary to improve upon what they receive.

That makes agricultural policy, at its core, an intergenerational undertaking.

The benefits extend well beyond agriculture itself. A productive and resilient agricultural sector contributes to economic activity, food security, scientific advancement, rural vitality, international trade, environmental stewardship, and the country's ability to withstand unexpected disruptions. Investments that strengthen agricultural capacity consequently serve interests that reach far beyond the farm gate.

Closing Statement

American agriculture has never been defined solely by the resources available to it. Its enduring strength has come from the ability to transform knowledge into innovation, innovation into greater productivity, and productivity into sustained economic and social benefits.

That process has depended upon institutions, communities, and individuals capable of learning from experience and adapting as circumstances change. The evidence reviewed throughout this assessment suggests that this capacity for adaptation remains one of the nation's greatest agricultural advantages.

The United States enters the coming decades with substantial strengths: productive land, extensive infrastructure, sophisticated scientific institutions, experienced producers, advanced technologies, developed markets, and generations of accumulated agricultural knowledge. The challenge is to ensure that these advantages continue to reinforce one another rather than gradually erode through neglect, fragmentation, or short-term decision-making.

Doing so will require patience. Many of the investments most important to agricultural capacity will not produce their full benefits within a single political term, budget cycle, or production

season. Their value may instead become apparent years or decades later, when future producers depend upon the research institutions, infrastructure, natural resources, knowledge, and human expertise that previous generations chose to preserve and strengthen. That is ultimately the measure of long-term agricultural policy.

The goal should not simply be to leave future generations the agricultural system that exists today. It should be to leave them a system with greater capacity to produce, innovate, adapt, and endure.

If the United States continues to invest in the scientific, human, environmental, economic, physical, and institutional foundations of agriculture, it will be better positioned to sustain a productive, resilient, and globally competitive agricultural sector throughout the next twenty-five to fifty years. The future cannot be predicted with certainty. But, the capacity to meet it can be built. That is the foundation of long-term agricultural success.

Bibliography

Alston, J. M., Andersen, M. A., James, J. S., & Pardey, P. G. (2010). *Persistence pays: U.S. agricultural productivity growth and the benefits from public R&D spending*. Springer.

Alston, J. M., Beddow, J. M., & Pardey, P. G. (2009). Agricultural research, productivity, and food prices in the long run. *Science*, 325(5945), 1209–1210.

<https://doi.org/10.1126/science.1170451>.

Fuglie, K., & Heisey, P. (2007). *Economic returns to public agricultural research* (Economic Brief No. 10). U.S. Department of Agriculture, Economic Research Service.

<https://doi.org/10.22004/ag.econ.6388>

Heisey, P., King, J., & Day-Rubenstein, K. (2011). Public research yields high returns... measured in more than dollars. *Amber Waves*. U.S. Department of Agriculture, Economic Research Service. <https://doi.org/10.22004/ag.econ.121098>

Heisey, P. W., Wang, S. L., & Fuglie, K. O. (2012). Public agricultural research investment helps determine productivity growth. *Amber Waves*. U.S. Department of Agriculture, Economic Research Service.

Huffman, W. E., & Evenson, R. E. (2006). *Science for agriculture: A long-term perspective* (2nd ed.). Blackwell Publishing.

Plastina, A., & Fulginiti, L. E. (2012). Rates of returns to public agricultural research in 48 U.S. states. *Journal of Productivity Analysis*, 37, 95–113. <https://doi.org/10.1007/s11123-011-0252-0>

Wang, S. L. (2020). *Productivity growth is the major source of growth in U.S. agricultural output*. USDA Economic Research Service, Charts of Note.

Wang, S. L., Ball, E., Fulginiti, L., & Plastina, A. (2012). Accounting for the impacts of public research, R&D spill-ins, extension, and roads in U.S. agricultural productivity growth. In K. O. Fuglie, S. L. Wang, & V. E. Ball (Eds.), *Productivity growth in agriculture: An international perspective* (pp. 13–31). CABI. <https://doi.org/10.1079/9781845939212.0013>

Wang, S. L., Heisey, P. W., Huffman, W. E., & Fuglie, K. O. (2013). Public R&D, private R&D, and U.S. agricultural productivity growth: Dynamic and long-run relationships. *American Journal of Agricultural Economics*, 95(5), 1287–1293. <https://doi.org/10.1093/ajae/aat032>

Wang, S. L., Heisey, P. W., Schimmelpfennig, D., & Ball, E. V. (2015). *Agricultural productivity growth in the United States: Measurement, trends, and drivers* (Economic Research Report No. 189). U.S. Department of Agriculture, Economic Research Service. www.ers.usda.gov/publications/err-economic-research-report/err189

Wang, S. L., Nehring, R., Mosheim, R., & Njuki, E. (2024). *Measurement of output, inputs, and total factor productivity in U.S. agricultural productivity accounts* (Technical Bulletin No. 1966). U.S. Department of Agriculture, Economic Research Service. <https://doi.org/10.22004/ag.econ.344773>

U.S. Department of Agriculture, Economic Research Service. (2026a). *Agricultural productivity in the United States*.

Dieter, C. A., Maupin, M. A., Caldwell, R. R., Harris, M. A., Ivahnenko, T. I., Lovelace, J. K., Barber, N. L., & Linsey, K. S. (2018). *Estimated use of water in the United States in 2015* (U.S. Geological Survey Circular 1441). U.S. Geological Survey. <https://doi.org/10.3133/cir1441>.

Wang, S. L., Hoppe, R. A., Hertz, T., & Xu, S. (2022). *Farm labor, human capital, and agricultural productivity in the United States* (Economic Research Report No. 302). U.S. Department of Agriculture, Economic Research Service. <https://doi.org/10.22004/ag.econ.327178>

Wang, S. L., & Njuki, E. (2025). *Quantity of agricultural labor has declined over time as quality of farm labor has grown*. USDA Economic Research Service, Charts of Note.

National Institute of Food and Agriculture. (2025). *Cooperative Extension System*. U.S. Department of Agriculture.

National Institute of Food and Agriculture. (2026). *Agriculture and Food Research Initiative—Education and Workforce Development*. U.S. Department of Agriculture.

National Academies of Sciences, Engineering, and Medicine. (2024). *Exploring linkages between soil health and human health*. National Academies Press. <https://doi.org/10.17226/27459>.

U.S. Department of Agriculture, Natural Resources Conservation Service. (2026a). *Soil health*.

U.S. Department of Agriculture, Natural Resources Conservation Service. (2026b). *Conservation Innovation Grants*.

U.S. Department of Agriculture, Climate Hubs. (2026a). *Ecosystem services*.

U.S. Department of Agriculture, Climate Hubs. (2026b). *Agriculture and a changing climate*.

U.S. Department of Agriculture, Forest Service. (n.d.). *Ecosystem services from National Grasslands*.

Hrozencik, R. A., Wallander, S., & Aillery, M. (2021). *Irrigation organizations: Water storage and delivery infrastructure* (Economic Bulletin No. 32). U.S. Department of Agriculture, Economic Research Service. <https://doi.org/10.22004/ag.econ.314931>

Lovelace, J. K., Nielsen, M. G., Read, A. L., Murphy, C. J., & Maupin, M. A. (2020). *Estimated groundwater withdrawals from principal aquifers in the United States, 2015* (U.S. Geological Survey Circular 1464). U.S. Geological Survey. <https://doi.org/10.3133/cir1464>.

Medalie, L., Galanter, A. E., Martinez, A. J., Archer, A. A., Luukkonen, C. L., Harris, M. A., & Haynes, J. V. (2025). *Water use across the conterminous United States, water years 2010–20* (U.S. Geological Survey Professional Paper 1894-D). U.S. Geological Survey. <https://doi.org/10.3133/pp1894D>.

U.S. Department of Agriculture, Agricultural Marketing Service. (2020). *The importance of highways to U.S. agriculture*.

U.S. Department of Agriculture, Agricultural Marketing Service. (2026). *Economic importance of inland waterways to U.S. agriculture*.

U.S. Department of Agriculture. (2019). *Next Generation Precision Agriculture & Broadband Report*.

McFadden, J., Njuki, E., & Griffin, T. (2023). *Precision agriculture in the digital era: Recent adoption on U.S. farms* (Economic Information Bulletin No. 248). U.S. Department of Agriculture, Economic Research Service.

U.S. Department of Agriculture, Agricultural Research Service. (2026a). *Long-Term Agroecosystem Research Network*.

U.S. Department of Agriculture, Agricultural Research Service. (2026b). *National Plant Germplasm System / Agricultural germplasm resources*.

Farrigan, T., Genetin, B., Sanders, A., Pender, J., Thomas, K. L., Winkler, R. L., & Cromartie, J. (2024). *Rural America at a glance: 2024 edition* (Economic Information Bulletin No. 282). U.S. Department of Agriculture, Economic Research Service.

<https://doi.org/10.32747/2024.8722498.ers>.

Lacy, K., Key, N., Bauman, A., Jablonski, B. B. R., & Hadrich, J. (2026). *Beginning farmer and rancher operations: Characteristics associated with business survival* (Economic Information Bulletin No. 294). U.S. Department of Agriculture, Economic Research Service.

<https://doi.org/10.22004/ag.econ.396401>

Baldwin, K. L., Miller, N., & Todd, J. E. (2026). *Risk management practices on U.S. farms and ranches, 1996–2020* (Economic Information Bulletin No. 296). U.S. Department of Agriculture, Economic Research Service. <https://doi.org/10.22004/ag.econ.396453>

U.S. Department of Agriculture, Economic Research Service. (2025a). *Risk management: Risk in agriculture*.

U.S. Department of Agriculture, Economic Research Service. (2025b). *Risk management: Risk management strategies*.

U.S. Department of Agriculture, Economic Research Service. (2026b). *Farm sector income & finances*.

Kreitman, T. (2026). *Deterioration in farm financial conditions remains gradual*. Federal Reserve Bank of Kansas City, Economic Bulletin, May 13, 2026.

Williams, A., Collins, L. A., & Boline, A. (2025). *Drivers of fertilizer markets: Supply, demand, and prices* (Economic Research Report No. 354). U.S. Department of Agriculture, Economic Research Service. <https://doi.org/10.22004/ag.econ.369107>

U.S. Department of Agriculture, Economic Research Service. (2026c). *Agricultural trade*.